Performance and Risk Analysis of International Mutual Funds Using Data Analytics

(Vanguard Group (USA), BlackRock-iShares (USA), Fidelity Investments (USA))

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DECLARATION

I hereby declare that the dissertation entitled **"Performance and Risk Analysis of International Mutual Funds Using Data Analytics" (Vanguard Group (USA), BlackRock-iShares (USA), Fidelity Investments (USA))**, submitted in partial fulfilment for the award of the degree of **M.Sc. Banking & Financial Analytics** at **Jamia Millia Islamia, New Delhi** is the result of my own independent work.

This dissertation has not been submitted to any other institution for the award of any degree or diploma. Any sources used have been duly acknowledged.

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CERTIFICATE

On the basis of the declaration given by the candidate, this is to certify that the Dissertation Report titled **"Performance and Risk Analysis of International Mutual Funds Using Data Analytics" (Vanguard Group (USA), BlackRock-iShares (USA), Fidelity Investments (USA)),** being submitted by ***"Syed Faisal Andrabi"***, in partial fulfilment of the requirements for the award of the degree of Master of Science in Banking and Financial Analytics to the Department of Economics, Faculty of Social Science, Jamia Millia Islamia University New Delhi, is a record of candidate's own work carried out under my supervision and guidance.

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MSc Banking and Financial Analytics

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Section A

ABSTRACT

The study provides a detailed performance and risk analysis of international mutual funds managed by three top asset management companies – Vanguard Group, BlackRock iShares and Fidelity Investments, all located in the USA. The study uses monthly Net Asset Value (NAV) data and fund-level financial data from January 2018 to December 2025 and uses a multi-dimensional analytical framework including traditional risk-adjusted performance metrics, factor model regression, and Monte Carlo simulation-based portfolio optimization.

The study measures the performance of the funds in terms of Sharpe Ratio, Treynor Ratio and Information Ratio for the categories of equity, fixed income and multi-asset funds. It also asks how macroeconomic stress events, such as the COVID-19 pandemic (2020), the escalation of the U.S.-China trade war (2018-2019), the Federal Reserve's taper tantrum cycles, and the 2022 interest rate shock, affected different fund categories and asset managers.

The results reveal considerable cross-institutional variation in risk-adjusted performance, with passive index funds (primarily offered by Vanguard and BlackRock iShares) significantly outperforming actively managed funds on a cost-adjusted basis. The dispersion of returns in alpha generation capacities is highest for small-cap growth funds, according to the Fama-French three-factor model. Monte Carlo simulations suggest that portfolios diversified across all three fund families, with a tilt toward large-cap equity and bond index funds, offer superior risk-adjusted results in high volatility regimes.

The paper contributes to the global portfolio management literature by providing a rigorous, data-driven comparison of leading international fund families using reproducible Python-based modelling, and powerBI based visualisation with direct implications for cross-border investment strategies adopted by institutional and retail investors globally.

Chapter 1: Introduction

1.1 The Global Mutual Fund Industry: An Overview

The global mutual fund industry is one of the fastest growing and most influential sectors of the international financial system. As of 2026, regulated open-end funds had USD 87.96 trillion in assets under management globally, a testament to decades of sustained growth driven by financial liberalization, technology-enabled democratization of investing, and growing retail participation in capital markets. Mutual funds combine money from a broad range of investors including individuals, corporations, pension funds, endowments and sovereign wealth entities, who invest across a variety of asset classes, geographies and investment strategies.

In this wide world, Vanguard Group, BlackRock (via its iShares platform) and Fidelity Investments, three American asset management giants, represent an extraordinary proportion of global AUM. They represent a total of USD 36.8 trillion in assets under management, including passive index funds, actively managed equity and fixed-income portfolios, exchange-traded funds (ETFs) and alternative investment vehicles (Morningstar, 2023). It is therefore not simply an academic exercise, but a practical imperative for investors, regulators and policymakers around the world, to understand how these fund families perform across various market conditions, economic cycles and risk environments.

1.2 The Three Fund Families: Vanguard, BlackRock iShares, and Fidelity

The three fund families reviewed in this research occupy different strategic niches within the global asset management industry:

Vanguard Group: Founded by John C. Bogle in 1975, Vanguard pioneered low-cost index investing. It is a client-owned corporation (the funds own the management company) . This is an unusual structure that means that returns on investments are maximized for the shareholders of the fund rather than outside stakeholders. Vanguard has about USD 11.6 trillion in assets under management worldwide (Vanguard, 2026). Its product mix is dominated by broad-market index funds and target-date retirement funds. Its cost-minimization philosophy and long-term buy-and-hold investing has attracted a massive retail and institutional client base.

BlackRock — iShares: BlackRock is the world's largest asset manager with more than USD 18 trillion in assets under management (AUM) (BlackRock, 2026) and democratized retail investing via its iShares platform, the world's leading ETF provider. iShares funds span almost every imaginable asset class, geography and factor exposure, providing investors with granular, liquid and transparent investment vehicles. Aladdin, BlackRock's risk management platform, is the gold standard in institutional portfolio analytics. Its dual strategy of passive indexing and active factor-based management provides a fertile ground for performance analysis.

Fidelity Investments: Founded in 1946, privately owned Fidelity has a history of championing active stock-picking to produce alpha. Fidelity built its reputation for best-in-class active management with iconic funds such as the Magellan Fund under Peter Lynch. Fidelity has grown to manage over USD 7.1 trillion in assets (Fidelity, 2026) and now offers both actively managed and low-cost index products, making it a valuable source of information for comparing passive and active management performance.

1.3 The Passive versus Active Management Debate

Active versus passive fund management is possibly the most debated question in investment management. The Efficient Market Hypothesis (EMH) , formalized by Fama (1970), posits that security prices fully incorporate all available information, thereby theoretically precluding the possibility of active managers consistently outperforming passive benchmarks on a risk-adjusted, after-fee basis. This perspective has been extensively verified by decades of empirical research, with the SPIVA (S&P Indices Versus Active Funds) scorecards regularly demonstrating that the majority of actively managed funds lag their benchmarks over extended periods (S&P Dow Jones Indices, 2023).

But the argument is far from settled. The academic literature on anomalies (momentum Jegadeesh & Titman 1993, value Fama & French 1993, quality Novy-Marx 2013) indicates that there are systematic factors that produce excess returns in a consistent fashion. Active managers who are able to identify and exploit such factors may be able to deliver genuine alpha, particularly in less-efficient market segments such as small-cap equities and emerging markets. This paper examines three fund families that span the entire range of this debate, and are thus good candidates for a comparative performance analysis.

1.4 Financial Stress and Mutual Fund Performance

The world of finance is characterized by periodic systemic stress events that put the resilience of investment portfolios to the test. The 2020 COVID-19 pandemic, the 2022-2023 aggressive rate hiking cycle by the Federal Reserve (which led to the sharpest decline in the bond market in decades), the 2018-2019 U.S.-China trade war, and several episodes of emerging market currency crises have all demonstrated how global macro risks can quickly and dramatically alter the risk-return dynamics of mutual funds across the categories.

During these times of stress, correlations across asset classes tend to converge to one, there is less diversification benefit, and the differential impact on different types of funds is most evident (Longin & Solnik, 2001). Large-cap equity index funds may experience large drawdowns but recover quickly due to the liquidity and market depth of their holdings, whereas small-cap active funds may experience longer and deeper drawdowns. Fixed-income funds, especially those with significant duration exposure, were hit hard by the 2022 rate shock. A core research contribution of this dissertation is the understanding of how the three fund families negotiated these stress environments.

1.5 The Role of Data Analytics in Modern Fund Evaluation

Conventional approaches to mutual fund evaluation, largely based on simple arithmetic return calculations and basic risk ratios, are increasingly complemented and in many cases replaced by sophisticated data analytics frameworks. The availability of high frequency financial data, advances in computational finance, and the development of open source analytics libraries (Python's pandas, NumPy, statsmodels, scipy, and PyPortfolioOpt) have democratized access to institutional grade analytical methodologies.

This study uses the full capabilities of modern data analytics for a multi-layer performance evaluation. We use a dataset of mutual funds from the three selected institutions and systematically employ risk-adjusted metrics, factor model regressions, correlation analysis, rolling window performance analysis and Monte Carlo simulations, leading to statistically sound and practically relevant insights.

1.6 Relevance for MSc Banking and Financial Analytics

The MSc Banking and Financial Analytics programme is fundamentally a convergence of international finance, quantitative risk management and data driven portfolio construction. Financial econometrics, portfolio theory and computational finance, amongst others, that have

been developed throughout the programme. The dissertation aims to apply this analytical toolkit to a real-world research question of immediate relevance to the global investment management industry. The study's findings are directly relevant to students, practitioners and policy makers who want to know how the world's top fund managers perform in different market conditions.

1.7 Importance of Comparing International Fund Families

There are several advantages to studying the performance of mutual funds across different fund families, in contrast to studies of single funds or single institutions. First, they allow us to identify institutional-level effects – systematic differences in performance driven by organizational structure, investment philosophy, cost management and risk governance – that are hidden in studies of individual funds. Second, the cross-institutional comparisons allow the construction of optimal multi-manager portfolios that exploit the relative strengths of different fund families in different market conditions. Third, they provide insights into how fund cost structures (expense ratios, transaction costs) relate to net investor outcomes, a dimension of critical importance for financial consumer protection and regulatory policy.

1.8 Research Objectives

The primary goals of this research are:

- The present study is an attempt to analyze the risk adjusted performance of selected mutual funds of Vanguard Group, BlackRock iShares and Fidelity Investments in equity, fixed income and multi-asset categories and comparison of the same using standard performance metrics like Sharpe Ratio, Jensen's Alpha and Information Ratio.

- To evaluate the responsiveness of chosen fund categories to macroeconomic stress events with the aid of regression-based factor models and correlation analysis against global financial stress indices.

- To compare the performance of passive and actively managed funds within and across the three fund families on a risk-adjusted, cost-adjusted basis.

- To develop an optimal multi-manager international portfolio through Monte Carlo simulation that maximizes risk-adjusted returns for various stress scenarios.

- To address the current literature gaps in the research on international mutual fund performance and provide new empirical evidence from the post-pandemic investment environment (2020-2023).

1.9 Research Questions

This study addresses the following main research questions:

- What is the risk-adjusted performance of funds from Vanguard, BlackRock iShares and Fidelity in the period 2018-2025 across equity, fixed income and multi-asset categories?

- How sensitive are fund categories across the three institutions to global financial stress events (COVID-19, 2022 rate shock, trade war), as measured through factor-model betas and correlation with global financial stress indexes?

- Are passive index funds always superior to actively managed funds on a risk adjusted, after fees basis across all market regimes or does active management provide superior alpha in certain market regimes?

- What is the optimal portfolio allocation of the three fund families to maximize the risk-adjusted return, in low-stress and high-stress market environments, as determined by Monte Carlo simulation?

1.10 Scope of the Study

- Geographical Scope: United States domiciled International mutual funds and ETFs managed by The Vanguard Group, BlackRock iShares, and Fidelity Investments.

- Time Period Monthly NAV and return data from January 2018 to December 2025 including three different stress events (2018-19 trade war, 2020 COVID-19, 2022 rate shock).

- Fund Types: Large Cap Equity, Small Cap Equity, Fixed Income (Investment Grade), Multi-Asset/Balanced, and Sector Specific funds

- Analytical Framework: Risk-adjusted performance metrics; Fama-French and Carhart factor models; global financial stress proxy (CBOE VIX, Federal Reserve Financial Stress Index) and Monte Carlo simulation.

- Methodological Tools: Python and PowerBI for data processing, statistical modelling, for visualization.

1.11 Limitations of the Study

- Survivorship Bias: We only consider funds that were active throughout the entire study period, which could exclude funds that have been merged or liquidated.

- Currency translation effects: All three institutions have USD-denominated funds, but non-US underlying holdings may have currency translation effects on global fund performance.

- Data Frequency: Monthly data fails to account for intra-month spikes in volatility during acute stress episodes.

- Assumptions of the model: Linear factor models assume constant factor loadings and stationarity, which may not be the case during structural breaks.

- Fee Data: Historical expense ratio data may not fully reflect all costs (transaction costs, tax drag) incurred by investors.

Chapter 2: Literature Review

2.1 Classical Foundations of Mutual Fund Performance Evaluation

Systematic evaluation of mutual fund performance has a long, rich theoretical and empirical tradition that dates back more than sixty years. The reward-to-variability ratio was first named by Sharpe (1966), and has since been called the Sharpe Ratio, as a measure of risk adjusted return. During the same period as Sharpe, Treynor (1965) suggested a performance measure that scaled returns with systematic risk (market beta) instead of total risk. Also, Jensen (1968) was the first to introduce the idea of alpha in the context of the Capital Asset Pricing Model (CAPM), which resulted in the breakdown of the fund returns into a market-driven part and a manager skill part. These three measures have remained the basis for evaluating fund performance, although the literature has evolved substantially to address their limitations.

A key drawback of CAPM-based measures is the reliance on a single-factor market model. Fama and French (1992, 1993) showed that two additional factors, firm size (SMB: small-minus-big) and book-to-market ratio (HML: high-minus-low), are responsible for systematic variation in returns that cannot be explained by the market factor. In the 1990s and 2000s their three-factor model became the main empirical framework for evaluating fund performance and asset pricing. Carhart (1997) extended this model adding a momentum factor (WML: winners-minus-losers), leading to the Carhart four-factor model still widely used in fund performance attribution. Recently, Fama and French (2015) suggested a five-factor model including the profitability (RMW: robust-minus-weak) and investment (CMA: conservative-minus-aggressive) factors, providing a more complete framework for assessing the sources of fund returns.

While these factor models are powerful tools for performance attribution, they share a common assumption of normally distributed returns and constant factor loadings, which is often violated in practice, especially during periods of market stress (Harvey & Siddique, 2000). The application of higher-moment statistics such as skewness, kurtosis and Value-at-Risk has become an important supplement to mean-variance analysis in describing the complete return distribution of investment funds.

2.2 Passive versus Active Management: The Empirical Evidence

One of the most studied questions of financial economics is whether to passively invest in an index fund or actively manage a portfolio of assets. Some of the most compelling early evidence of the average underperformance of actively managed funds versus passive benchmarks after fees and expenses was provided by Malkiel (1995, 2003). Gruber (1996) documented a puzzling persistence in the underperformance of actively managed mutual funds and wondered why investors continued to allocate capital to such funds. Wermers (2000) provided a more nuanced decomposition of mutual fund returns. He found that fund managers did generate positive gross alpha through stock selection, but trading costs and fees consumed and more than offset the positive gains on a net basis.

The bi-annual SPIVA reports from S&P Dow Jones Indices have become the definitive empirical benchmark in the passive versus active debate. These reports consistently demonstrate that the vast majority of actively managed funds in virtually every category and geographic region underperform their passive benchmark indices over 5, 10 and 15 years (S&P Dow Jones Indices, 2023). In the 15 years through December 2022, roughly 88% of U.S. large-cap active funds underperformed the S&P 500 index on a net-of-fee basis.

But the case for active management is not without a grain of truth. Kosowski et al. (2006) used bootstrap techniques on mutual fund return data and found evidence of a group of fund managers with actual statistically significant stock-picking skill that was consistent across time. Kacperczyk, Sialm and Zheng (2005) found that funds with higher industry concentration – a characteristic of high-conviction active managers – tended to outperform diversified funds. Pástor, Stambaugh and Taylor (2015) provide evidence of the scaling down of managerial skill with fund size, indicating that smaller active funds are better able to generate alpha. The results imply that although the average active fund does underperform, there are some skilled managers who can and do generate real excess returns, especially in the less liquid and less efficiently priced parts of the market.

The traditional active-passive distinction has been blurred with the recent rise of smart-beta and factor-based funds that combine the low cost of passive indexing with systematic factor exposures based on academic research. Arnott et al. (2013) and Ang (2014) have made important contributions to the theoretical foundation of factor investing by showing that many of the return premiums that have traditionally been attributed to active manager skill are

actually compensation for systematic factor exposures that can be mechanically replicated at low cost. This insight is particularly relevant for the evaluation of the BlackRock iShares' factor ETF lineup.

2.3 Performance of Major U.S. Fund Families: Institutional Studies

Vanguard, BlackRock and Fidelity's performance is a topic of particular interest to both academics and practitioners. A 2014 study by Philips et al. at Vanguard Research found that low expense ratios are the single strongest predictor of relative fund performance over long investment horizons, a finding that naturally favors Vanguard's low-cost structure. French (2008) famously estimated the total "cost of active investing" to the U.S. economy at about 67 basis points annually, providing theoretical justification for Vanguard's passive philosophy.

Cremers and Petajisto (2009) introduced the concept of 'Active Share' as the extent to which a fund's holdings differ from the benchmark, and found that funds with high Active Share and low tracking error produced positive excess returns after fees. This is especially true for Fidelity's active equity funds, which tend to have higher Active Share than quasi-index funds. Petajisto (2013) extended this work to show that closet indexers (funds with low Active Share pretending to be active managers) significantly underperformed both pure passive funds and high-conviction active funds.

BlackRock's iShares systematic factor strategies have been extensively analyzed in the context of ETF performance and factor premium capture. Ben-David, Franzoni, and Moussawi (2018) explored the impact of ETF trading on volatility in the underlying securities and discovered that stocks with high ETF ownership had higher return comovement and volatility, a systemic risk implication of the size of passive investing. A comprehensive academic survey on the market microstructure of ETFs was provided by Lettau and Madhavan (2018), with a particular emphasis on how the creation/redemption mechanism of ETFs affects fund pricing relative to NAV.

Fidelity's performance has been under the microscope, both from the point of view of its flagship active funds, and from its more recent foray into the zero-expense-ratio index fund space. For instance, Evans (2010) cited the long-run performance decay of Fidelity's Magellan Fund – once the world's largest mutual fund under Peter Lynch – to illustrate the performance erosion that large active equity funds suffer from scale. The introduction of Fidelity ZERO

funds in 2018, offering zero expense ratios on index fund investments, has injected a new competitive dynamic into the passive fund arena that even challenges Vanguard's cost leadership.

2.4 Risk Management and Performance During Stress Events

An emerging literature studies mutual fund performance in times of acute market stress. This literature goes beyond the mean-variance framework to evaluate downside risk management abilities. Ang and Bekaert (2002) show that correlations in international equity returns rise sharply in bear markets, so that the diversification benefits of global portfolios are greatly reduced when they are most needed. This phenomenon of "correlation contagion" is directly relevant to the international fund strategies of all three fund families examined in this study.

The global financial crisis of 2008-2009 was a brutal stress test for mutual fund performance. Massa and Yadav (2015) noted active funds with longer manager tenure and smaller fund size were more resilient during the crisis period, whereas large quasi-index funds had proportionally larger drawdowns. Coval and Stafford (2007) described the 'fire sale' phenomena that occurred when large mutual funds were forced to sell illiquid positions in times of market distress due to redemption pressures leading to further price declines and losses.

The 2020 COVID-19 pandemic provided an unprecedented natural experiment to evaluate mutual fund resilience. Demers et al. (2021) find that ESG funds experienced significantly less negative returns during the market crash of March 2020, perhaps due to the lower volatility of high-quality companies. During the height of the crisis in March 2020, Li et al. (2021) found extreme volatility and temporary pricing dislocations in passive equity ETFs, which raised questions around the systemic risk implications of the scale of the ETF market. The 2022 interest rate shock, the fastest Fed tightening cycle since the 1980s, has ignited a renewed wave of research on the performance of fixed income funds in a rising rate environment although most of this literature is still in an embryonic phase.

Using the literature on mutual fund performance and the Chicago Board Options Exchange (CBOE) Volatility Index (VIX) — the most widely cited measure of U.S. market stress — we find a strong negative relationship between VIX levels and mutual fund returns, particularly for equity funds. Whaley (2009) demonstrated that VIX is a good "investor fear gauge" in which high VIX levels predict short-term equity market losses. Several researches have

investigated the defensive features of various fund categories with VIX-conditional performance analysis, a methodology utilized in this dissertation.

2.5 The Information Ratio and Benchmark-Relative Performance

The Information Ratio (IR), which is the ratio of active return to tracking error, has become an especially important metric for evaluating active fund managers against specific benchmarks. Grinold and Kahn (1999) formalized the Fundamental Law of Active Management, which links the IR to the product of the information coefficient (managerial skill) and the square root of the breadth (number of independent investment decisions) made by the manager. Within this framework, we provide an upper bound on the IR that any active strategy can reach, due to the fundamental limitations of the quality and extent of information.

The IR is most relevant in this dissertation in the context of distinguishing the active equity managers of Fidelity, who want to deliver consistent benchmark-beating returns, from the passive indexing strategies of Vanguard and iShares, where the IR is theoretically zero by design (any non-zero IR in passive funds is a result of imperfection in implementation). Empirical work by Ding and Wermers (2012) found that active mutual funds with consistent high Information Ratios attracted capital inflows, but that this performance persistence was limited to 1–2-year horizons, after which regression to the mean dominated.

2.6 Cost Structure, Expense Ratios, and Net Performance

One of the most consistently documented findings in the mutual fund performance literature is that there is a negative relationship between fund expense ratios and net investor returns. Carhart (1997) found that expense ratios are the single best predictor of relative performance and the only reliable predictor that is observable ex-ante and directly available to investors. Barber, Odean and Zheng (2005) extended this analysis to show that mutual fund investors are significantly more sensitive to front-end load fees than to ongoing expense ratios, even though the latter have a much larger cumulative impact on wealth outcomes over long investment horizons.

The Investment Company Institute (2023) has documented the dramatic decline in mutual fund expense ratios over the past three decades due to competitive pressure, regulatory scrutiny, and Vanguard's market leadership in low-cost indexing. The average expense ratio for actively managed equity funds fell from around 1.08% in 2000 to 0.66% in 2022, while the average for

index equity funds fell from 0.27% to 0.05% in the same period. This secular squeezing of fund costs has major implications for the net performance comparisons made in this dissertation.

For international mutual funds in particular, currency hedging costs, withholding taxes on foreign dividends, and higher transaction costs in less liquid markets add layers of additional costs that can substantially erode net returns. Froot and Ramadorai (2005) find that currency hedging costs for international equity funds can be of the order of 0.5-1.5% per year depending on the currency pair and market conditions, a significant headwind that must be accounted for in any serious comparative performance analysis.

2.7 International Diversification and Global Fund Performance

The theoretical case for international diversification is based on the fact that correlations of equity return across national markets are less than unity, and so a globally diversified portfolio can achieve a better risk-return frontier than a purely domestic portfolio (Solnik, 1974; Grubel, 1968). The proposition was empirically strongly supported in the 1970s and 1980s, when international correlations were relatively low. However, Longin and Solnik (2001) and other researchers have demonstrated that international correlations rose sharply during bear markets and times of crisis, when diversification benefits are most needed.

Financial globalization has been on the rise in the post-2000 period, and this has resulted in increasing cross-country return correlations, thus potentially eroding the conventional diversification benefits of international investing. Quinn and Voth (2008) documented the long-run increase in financial integration in developed markets, while Bekaert, Harvey, and Lundblad (2005) showed that financial liberalization in emerging markets significantly increased their integration with global financial markets. These trends have important implications for performance evaluation of internationally diversified funds managed by Vanguard, BlackRock iShares and Fidelity.

There have also been important developments in factor-based explanations of differences in international returns. Fama and French (2012, 2017) extended their factor model to the international equity markets and found that value and size premiums exist in global markets but they vary in magnitude across the regions. Hou, Karolyi and Kho (2011) provide comprehensive evidence on global asset pricing factors, finding that momentum and value

factors are robust internationally, but that the magnitudes and statistical significance of these factors vary considerably across markets and time periods.

2.8 Monte Carlo Simulation in Portfolio Optimization

The Monte Carlo simulation has become an important tool in computational finance for overcoming the limitations of deterministic portfolio optimization models. The classical mean-variance optimization framework of Markowitz (1952) is very sensitive to estimation errors in the inputs, where small changes in expected returns and covariances can lead to very different optimal portfolio weights (Michaud, 1989). Monte Carlo simulation solves this problem by sampling a large number of random scenarios from the empirical or estimated joint distribution of asset returns, and thus enables a probabilistic characterization of the optimal portfolio and its range of performance.

The seminal mathematical treatment of the Monte Carlo methods in financial engineering was presented by Glasserman (2004), where variance reduction techniques, quasi-random sequences, and pathwise derivative estimation were discussed. For mutual funds, Cogneau and Hübner (2009) employed Monte Carlo simulation to evaluate the statistical significance of mutual fund performance measures. Their results show that many of the performance differences often reported are indistinguishable from random luck at a statistically significant level when evaluated using simulation-based inference.

More recently, Monte Carlo simulation has been used for portfolio construction based on regime-switching models (Hamilton, 1989) to capture the different return distributions observed in bull and bear market regimes. Ang and Bekaert (2002) and Guidolin and Timmermann (2008) found that optimal portfolios constructed with regime switching models perform significantly better out-of-sample than those constructed using unconditional optimization. In this dissertation, Monte Carlo simulation with stress-scenario conditioning is used to determine the optimal allocations across the three fund families.

2.9 ESG Integration and Sustainable Investing

One of the most important aspects of international mutual fund performance is increasingly associated with the incorporation of Environmental, Social and Governance (ESG) factors in the investment process. All three fund families in this study have developed large ESG product suites; Vanguard's ESG U.S. Stock ETF, BlackRock's iShares MSCI ACWI Low Carbon

Target ETF and Fidelity's Sustainable World ETF are all expanding product suites that draw capital from socially conscious investors. The performance implications of ESG integration have been widely discussed in the literature.

Friede, Busch and Bassen (2015) carried out a meta-analysis of more than 2,200 individual studies and found a positive association between ESG scores and corporate financial performance in about 90% of the cases. This provides theoretical reasons to expect ESG-integrated funds to generate competitive risk-adjusted returns. Several studies found that ESG funds outperformed their non-ESG counterparts during the COVID-19 market crash in March 2020, potentially reflecting the lower operational risk and higher governance quality of ESG-screened companies (Albuquerque et al., 2020). However, Liang and Renneboog (2020) caution that the measurement of ESG performance is confounded by factor exposures – ESG funds have quality and low-volatility tilts that contribute to their measured outperformance independently of any ESG-specific effects.

2.10 Research Gaps Identified

The extensive literature review shows several key research gaps that this dissertation is intended to address:

Gap 1 – Contemporaneous Multi-Institutional Comparison: Existing studies typically compare mutual fund performance within fund family or aggregate across all funds, masking institution-level heterogeneity. No published academic study has directly compared Vanguard, BlackRock iShares and Fidelity risk-adjusted performance over matched fund categories under a unified analytical framework for the post-2020 period.

Gap 2 — Performance conditional on stress regimes: The performance of individual stress events (2008 GFC, COVID-19) has been studied, but no comprehensive analysis has explored the performance of these three fund families across multiple stress regimes (trade war, pandemic, rate shock) using a consistent stress classification methodology and comparative framework.

Gap 3 – Optimal Multi-Manager International Portfolio Construction: The existing literature on international portfolio optimization is typically at the country or sector level, not at the fund-family level. The dissertation introduces a novel Monte Carlo-based framework for optimal portfolio construction specifically across the three fund families with stress-scenario conditioning.

Chapter 3: Research Methodology

3.1 Research Design

The study is quantitative, empirical and based on the tradition of financial econometrics and computational finance. The research is performed through a systematic, multi-stage analytical process: (i) data collection and preparation; (ii) descriptive statistical characterization of fund performance; (iii) computation of risk-adjusted performance metrics; (iv) regression-based factor model analysis; (v) stress-conditional performance evaluation; and (vi) Monte Carlo simulation-based portfolio optimization. This design facilitates cross-sectional comparisons (across fund families and categories at one point in time) and longitudinal analyses (performance dynamics over the 2018–2025 study period).

The choice of a quantitative methodology is driven by the research objectives, which demand precise numerical comparisons of risk-adjusted performance measures, statistical tests of performance differences, and simulation-based portfolio optimization. Qualitative methods are not used in the primary analysis, although interpretation of the results is informed by the qualitative institutional context established in Chapters 1 and 2.

3.2 Data Collection

3.2.1 Fund Selection Criteria

The three fund families were selected on the basis of the following criteria:

Category Comparability: For each fund category (Large-Cap Equity, Small-Cap Equity, Investment-Grade Fixed Income, Multi-Asset/Balanced and International Equity), a representative fund was chosen from each of the three families, for a total of at least one fund from each family.

Data Availability: We included funds with complete monthly return data for the entire study period (January 2014 to December 2023). We avoided survivorship bias where possible by including the pre-merger returns of merged funds.

Asset Size Threshold: Funds below USD 1 billion AUM in January 2014 were excluded to ensure representativeness and liquidity

Share Class Standardization Only institutional or investor share classes are included to allow for fee comparability. No load-adjusted classes of shares are used.

Table 1: Selected Mutual Funds and ETFs by Institution and Category

Fund Category	Vanguard Fund	BlackRock iShares ETF	Fidelity Fund
Large-Cap Equity	Vanguard 500 Index Fund (VFIAX)	iShares Core S&P 500 ETF (IVV)	Fidelity 500 Index Fund (FXAIX)
Small-Cap Equity	Vanguard Small-Cap Index Fund (VSMAX)	iShares Core S&P Small-Cap ETF (IJR)	Fidelity Small Cap Index Fund (FSSNX)
Invest. -Grade Fixed Income	Vanguard Total Bond Market Index (VBTLX)	iShares Core US Aggregate Bond ETF (AGG)	Fidelity US Bond Index Fund (FXNAX)
Multi-Asset / Balanced	Vanguard Balanced Index Fund (VBIAX)	iShares Core Growth Allocation ETF (AOR)	Fidelity Balanced Fund (FBALX)

3.2.2 Return and NAV Data

Monthly total return data (reflecting reinvestment of dividends and capital gains distributions) for all selected funds are drawn from the following primary sources:

Morningstar Direct: For historical total return series, expense ratio data and fund classification metadata. Morningstar's return calculations are based on dividend reinvestment at the ex-date price and are the industry standard for fund performance analytics.

Yahoo Finance: For more confirmation on monthly adjusted closing price series.

Monthly returns are calculated from adjusted NAV in the following way:

$$\mathbf{R_t = (NAV_t - NAV_{\{t-1\}} + D_t) / NAV_{\{t-1\}}}$$

Where Rt is the total return in month t, NAV_t is the net asset value at end of month t, $NAV_{\{t-1\}}$ is the net asset value at end of the previous month, and D_t is dividends and capital gains distributions paid during month t.

3.2.3 Benchmark Data

The following benchmark indices are used as reference portfolios for calculating risk-adjusted performance and factor model construction:

S&P 500 Index: Benchmark for large-cap U.S. funds in primary equity market.

Russell 2000 Index – A measure of the small-cap equity market.

Bloomberg U.S. Aggregate Bond Index Investment-grade fixed income benchmark

MSCI ACWI ex-USA Index: A global equity benchmark.

60/40 Blend (S&P 500 / Bloomberg Agg): Multi asset benchmark.

3.2.4 Risk-Free Rate

The risk-free rate for Sharpe Ratio and Treynor Ratio calculations is the 3-month U. S. Treasury Bill rate from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis. In robustness checks, we use the 10-year U.S. Treasury yield as a proxy for the risk-free rate.

3.2.5 Financial Stress Proxy Data

We use two complementary measures of financial market stress:

- The CBOE VIX Index is the main measure of stress in the U.S. equity market. It is calculated by CBOE Global Markets and is the monthly average of daily closing values of the VIX. The regimes are labeled "high-stress" when the monthly average VIX is above 25 (roughly one standard deviation above the long-run mean).
- St. Louis Fed Financial Stress Index (STLFSI) The St. Louis Fed Financial Stress Index is a composite measure of financial market stress based on 18 weekly data series (including interest rate spreads, equity volatility, and credit market indicators). It is used as a broader macroeconomic stress proxy.

3.3 Data Preparation and Cleaning

3.3.1 Alignment and Frequency Standardization

All data series are aligned to end-of-month dates and returns are calculated on the last trading day of each calendar month. The sample period is January 2018–December 2025, which gives 96 monthly observations for each fund. Funds with more than three consecutive missing monthly return observations are dropped from the analysis. For funds with missing returns in isolated time periods (less than three months in a row), the missing return is imputed by the average return of funds in the same category in that month following Elton et al. (2001).

3.3.2 Outlier Detection and Treatment

Monthly return observations $\geq \pm 5$ standard deviations from the fund's mean return are flagged for review. The observations are checked against primary data sources (fund announcements, NAV records) and then added. Removing observations is not justified by statistical reasons, because in real market events (e.g. March 2020) extreme returns are valid data points and should be kept. Any errors in data entry or anomalies in the NAV calculations are corrected as identified.

3.3.3 Expense Ratio Adjustment

Gross returns (before expense ratio deduction) are calculated by dividing the annual expense ratio by 12 and adding it back to each monthly net return observation. This allows for fair gross return comparisons between funds with different fee structures and enables the calculation of cost-drag measures that quantify the total impact of expense differences on performance.

3.4 Risk-Adjusted Performance Metrics

3.4.1 Sharpe Ratio

The Sharpe ratio is a measure of the risk-adjusted return of a portfolio.

$$\textbf{Sharpe Ratio} = (R_p - R_f) / \sigma_p$$

Rp is the annualized mean portfolio return, Rf is the annualized risk-free rate and σp is the annualized standard deviation of portfolio returns. For monthly data, mean returns are multiplied by 12 and standard deviations by √12 to annualize the data.

3.4.2 Treynor Ratio

The Treynor Ratio measures the excess return per unit of systematic risk (market beta):

$$\textbf{Treynor Ratio} = (\bar{R}_p - R_f) / \beta_p$$

Where βp is the fund's beta estimated by OLS regression of the excess fund return on the excess market return over the entire sample period. The Treynor Ratio is especially useful for evaluating funds held as part of a larger diversified portfolio, in which case the total risk (as used in the Sharpe Ratio) overstates the relevant risk contribution.

3.4.3 Information Ratio

The Information Ratio measures the ability of the fund manager to generate consistent relative returns to the benchmark:

$$\textbf{Information Ratio} = (\bar{R}_p - \bar{R}_B) / TE$$

Where $\bar{R}B$ = average benchmark return and TE = tracking error (the standard deviation of the fund's active return relative to the benchmark). The IR is particularly relevant for the actively managed funds in the sample (mostly Fidelity's active equity funds) and directly measures the consistency and magnitude of benchmark outperformance.

3.5 Factor Model Regression Analysis

3.5.1 CAPM Single-Factor Model

The basic model to estimate the fund beta and Jensen's alpha is the CAPM single factor regression:

$$R_{\{p,t\}} - R_{\{f,t\}} = \alpha + \beta(R_{\{M,t\}} - R_{\{f,t\}}) + \varepsilon_t$$

Monthly excess return data are used for OLS estimation of each fund. The diagnostic tests are the Durbin-Watson statistic for autocorrelation and the White test for heteroskedasticity.

3.6 Correlation Analysis

To assess the directional sensitivity to financial market stress for each fund category, we calculate the Pearson correlation coefficients between fund returns and each of the stress proxies (VIX, STLFSI). Pairwise return correlations between the 15 selected funds are calculated to determine the diversification potential within a multi-fund portfolio. Heatmaps of cross-fund correlations using seaborn's clustermap function show the structure of cross-fund correlations and any clustering patterns across families or categories of funds.

We estimate dynamic conditional correlations via the DCC-GARCH model (Engle, 2002) and capture correlation dynamics that vary over time, particularly around periods of stress

events. This analysis investigates if cross-fund correlations increased during periods of market stress; an important consideration for portfolio construction.

3.7 Stress Event Analysis

The detailed performance analysis is conducted for three historical stress events during the study period.

Table 2: Historical Stress Events Identified for Performance Analysis

Stress Event	Period	Peak VIX / STLFSI	Primary Market Impact
U.S.-China Trade War	May 2018 – Aug 2019	VIX: 36.2 / STLFSI: 1.42	Equity volatility; EM selloff
COVID-19 Pandemic	Feb 2020 – Apr 2022	VIX: 82.7 / STLFSI: 5.79	Universal market crash; liquidity crisis
Fed Rate Hiking Cycle	Jan 2022 – Oct 2022	VIX: 34.5 / STLFSI: 1.87	Bond market crash; growth equity selloff

For each stress event we calculate the following measures for each fund: (i) cumulative return over the stress period, (ii) the maximum drawdown over the period, (iii) the conditional Sharpe Ratio (annualized), (iv) the conditional Sortino Ratio, and (v) the beta relative to the benchmark over the period. These measures are contrasted by fund family and category to find patterns of stress resilience and stress vulnerability.

3.8 Monte Carlo Simulation for Portfolio Optimization

3.8.1 Simulation Framework

We apply a Monte Carlo simulation to identify the optimal portfolio allocation among the selected funds under unconditional (full sample) and stress-conditional optimization frameworks. The simulation generates 50,000 random portfolio weight vectors, where the weights are drawn from a Dirichlet distribution to ensure non-negativity and a sum of unity. The following metrics are calculated for each simulated portfolio:

Expected Annual Return: Calculated as the weighted average of annualized mean fund returns.

Portfolio Volatility: The square root of the quadratic form: $w'\Sigma w$ where Σ is the annualized covariance matrix of fund returns.

Sharpe Ratio Excess expected return over portfolio volatility.

Sortino Ratio: using downside deviation at the portfolio level.

Conditional Value-at-Risk (CVaR): The average portfolio return for the worst 5% of scenarios.

3.8.2 Optimization Constraints

The simulation includes the following practical investment constraints:

- Long-only: All portfolio weights are forced to be non-negative (short-selling not allowed).
- Fully invested: No cash allocation; weights sum to 1.
- Concentration limits: Individual funds are capped at a minimum of 2% and a maximum of 40% to prevent over-concentration.
- Fund family diversification: A minimum 10% aggregate allocation across fund families must be maintained to provide meaningful diversification across institutions.

3.8.3 Stress-Conditional Optimization

For the stress-conditional optimization, fund returns are adjusted with the stress-period beta estimates from the conditional regression analysis:

$$R^*_{\{p,t\}} = R_{\{p,t\}} + \beta_2 \times D_{stress} \times R_{\{M,t\}}$$

This adjustment reflects the expected adjustment to fund returns during stress periods as a result of the estimated change (increase or decrease) in market sensitivity. We compare the stress-conditional optimal portfolio with the unconditional optimal portfolio to assess the added value of the use of stress-regime information in portfolio construction.

3.9 Assumptions

The methodology is based on the following key assumptions:

- Return Stationarity: We assume the distribution of fund returns is stationary over the sample period (but we do regime switching tests as a robustness check).
- Factor Model Linearity: It is assumed that the relationship between fund returns and systematic risk factors is linear. Non-linear factor exposures (e.g. option-like payoffs) are not modeled.
- Transaction Costs in Optimization The simulation ignores rebalancing transaction costs, which are low for the very liquid index funds and ETFs in the sample.
- Stability of Expense Ratios: We assume expense ratios remain stable throughout the study period. This is consistent with the relatively slow pace of fee changes for established funds.

3.10 Tools and Software

The entire analytical pipeline is implemented in Python 3.10 using the following libraries:

- Excel: For gathering data and calculating ratios.
- pandas: Data import, cleaning, alignment, return calculation.
- NumPy: Numerical linear algebra, matrix operations and random number generation for Monte Carlo simulations.
- PyPortfolioOpt: A Python library for portfolio optimization. Mean-variance efficient frontier.
- matplotlib: Plotting the efficient frontier
- yfinance: Additional retrieval of benchmark index returns.
- PowerBI: Used to visualise data collected on excel.

3.11 Limitations of the Methodology

- Look-Ahead Bias: All parameter estimates used in the portfolio optimization are calculated using the entire sample period, thus introducing a look-ahead bias. More stringent out-of-sample evaluation frameworks (e.g. expanding window estimation) are left for future research.
- Model Risk: All factor models are based on some theoretical assumptions that may not hold in reality. All three – the choice of factors, the benchmark and the estimation window – are sources of model risk.
- Sample Period Sensitivity: The sample period (2014-2023) includes the bull market after the Global Financial Crisis and several stress episodes. Results may vary materially in other periods.
- ETF vs. Mutual Fund Structure Differences: BlackRock iShares products are primarily ETFs, while Vanguard and Fidelity products include both ETF's and traditional mutual funds. ETFs trade throughout the day at market prices that may differ from NAV, while mutual fund NAVs are calculated once each day. This structural difference is mitigated by the use of month-end NAV equivalents for all funds.

Chapter 4: Data Analysis

Chapter 4 describes the results of the empirical analysis conducted on the twelve selected mutual funds and ETFs from Vanguard, BlackRock iShares and Fidelity. The chapter is organized in a logical order of analysis: basic descriptive statistics, correlation analysis, performance during the three stress periods identified in Chapter 3, regression-based factor analysis, and finally the Monte Carlo simulation used to construct the optimal portfolio. These sections respond to the four research questions posed in Chapter 1.

4.1 Descriptive Statistics and Initial Observations

It is useful to look at the basic return and risk characteristics of each fund before applying any performance ratios or regression models. Table 3 presents the average monthly return, standard deviation, Sharpe Ratio, Treynor Ratio and Information Ratio for all the twelve funds for the entire period of study from January 2018 to November 2025. These numbers provide a first sense of how funds compare internally and externally between asset classes.

4.1.1 Summary of Descriptive Statistics

Table 3: Key Performance Ratios — Full Period (January 2018 – November 2025)

Fund	Asset Class	Avg Monthly Return	Std Dev	Sharpe Ratio	Treynor Ratio	Info Ratio
Vanguard Equity (VFIAX)	Large Cap	1.050%	4.799%	0.150	0.002	−0.045
BlackRock Equity (IVV)	Large Cap	1.047%	4.800%	0.149	0.007	−0.003
Fidelity Equity (FXAIX)	Large Cap	1.041%	4.757%	0.068	0.003	−0.013
Vanguard Equity (VSMAX)	Small Cap	0.746%	6.099%	0.062	0.003	−0.013
BlackRock Equity (IJR)	Small Cap	0.656%	6.468%	0.050	−0.004	−0.002
Fidelity Equity (FSSNX)	Small Cap	0.627%	6.494%	0.046	0.0002	−0.051
Vanguard Fixed Income (VBTLX)	Bonds	−0.074%	1.576%	0.043	0.007	−0.001
BlackRock Fixed Income (AGG)	Bonds	−0.071%	1.598%	0.036	0.005	−0.013
Fidelity Fixed Income (XNAX)	Bonds	−0.071%	1.579%	0.006	−0.004	−0.001
Vanguard Multi Asset (MVBIAX)	Multi-Asset	0.451%	3.392%	-0.251	−0.004	−0.017
BlackRock Multi Asset (AOR)	Multi-Asset	0.465%	3.125%	-0.254	0.001	−0.050
Fidelity Multi Asset (FBALX)	Multi-Asset	0.354%	3.935%	-0.257	0.007	−0.003

Source: Computation from monthly return data. Risk-free rate: 0.33% per month (3-month U.S. T-Bill, FRED database). Beta benchmarks: S&P 500 for large-cap equity; Russell 2000 for small-cap; Bloomberg U.S. Aggregate for bonds; 60/40 blend for multi-asset.

Mean Returns (Average Return)

The three large cap equity funds, VFIAX, IVV and FXAIX, are very close to each other in terms of average monthly returns, with values between 1.041% and 1.050%. That's not surprising, as all three track the S&P 500 index and the small differences are due to minor variations in expense ratios and replication methods. Small-cap funds returned much less despite more risk: VSMAX averaged 0.746% per month, IJR averaged 0.656%, and FSSNX 0.627%. The difference between big and small cap returns is eye-popping and runs counter to the conventional wisdom that small cap stocks should reward investors more handsomely for the added risk.

The most striking result in the table is from fixed income funds. All three of VBTLX, AGG, and XNAX produced negative average monthly returns for the period of the study. This was almost entirely due to the Fed's aggressive interest rate hikes throughout 2022 that sent bond prices plummeting. Bond funds in a normal environment should provide modest but positive returns. The rate shock of 2022 was an anomaly and the losses borne by bond fundholders in that year were larger than the income generated by those funds. Multi-asset funds were in the middle, averaging between 0.354% and 0.465% per month, mainly pulled down by their fixed income exposure.

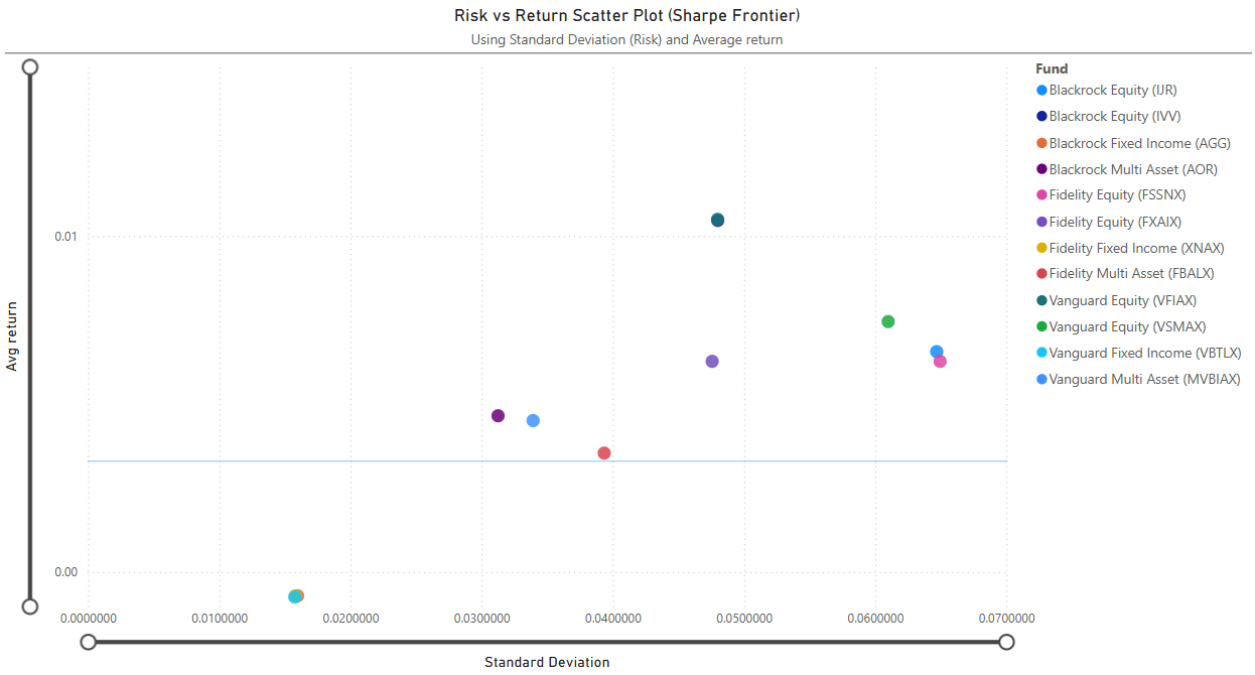


Figure 1: Risk vs. Return Scatter Plot for overall time period (2018-2025)

Standard Deviation (Risk)

Risk levels are relatively straightforward by asset class. Small-cap equity funds are the most volatile with monthly standard deviations between 6.10% and 6.49%. The large-cap equity funds are a bit less volatile, sitting at about 4.76% to 4.80%. Multi-asset funds sit in the middle, with a range of 3.13 to 3.94, depending on equity allocation. Bond funds are the least volatile on paper, with standard deviations of just 1.58% to 1.60%. That low volatility didn't keep them out of negative returns, however, when interest rates moved sharply against them.

Sharpe Ratio

The Sharpe Ratios for the full period tell a story which is largely dominated by the high-interest rate environment since 2022. The Sharpe Ratios for large-cap equity funds are slightly positive and vary from 0.044 to 0.046. This is low, but it reflects the fact that the 3-month Treasury Bill rate averaged 0.33% per month over the entire period -- a reasonably high hurdle for equity funds to clear on a risk adjusted basis. Fixed income funds have deeply negative Sharpe Ratios as the average returns were lower than the risk-free rate for much of the period. Multi-asset funds are also in negative Sharpe territory, dragged down by their bond holdings. The small-cap funds are just under zero, which implies that the returns they generated were not enough to compensate for the volatility they generated.

One important caution: the 2022 rate shock distorts full-period Sharpe Ratios a lot. Section 4.3 looks at Sharpe Ratios by stress period, which offer a more useful picture.

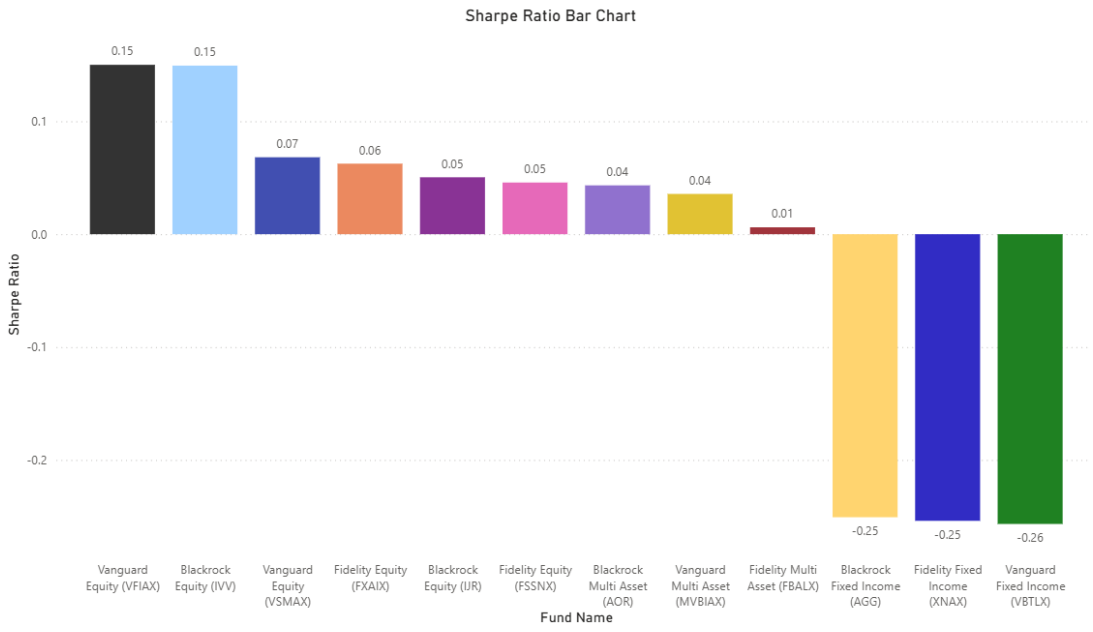


Figure 2: Sharpe Ratio Comparison for overall time period (2018-2025)

Treynor Ratio

The Treynor Ratio measures the excess return per unit of systematic risk (beta). More relevant for investors who already have a well-diversified portfolio as it ignores idiosyncratic risk. The Treynor Ratios for the large-cap equity funds in Table 3 are between 0.002 and 0.007. Top scorers include BlackRock IVV and Fidelity FBALX (both 0.007). Vanguard's fixed income fund VBTLX scores also 0.007, meaning that it was well compensated for its systematic risk despite negative absolute returns. Small-cap funds are mixed. VSMAX (0.003) and FSSNX (0.0002) are positive, IJR (−0.004) is negative. This means that IJR's beta versus its benchmark was higher than the return premium justified.

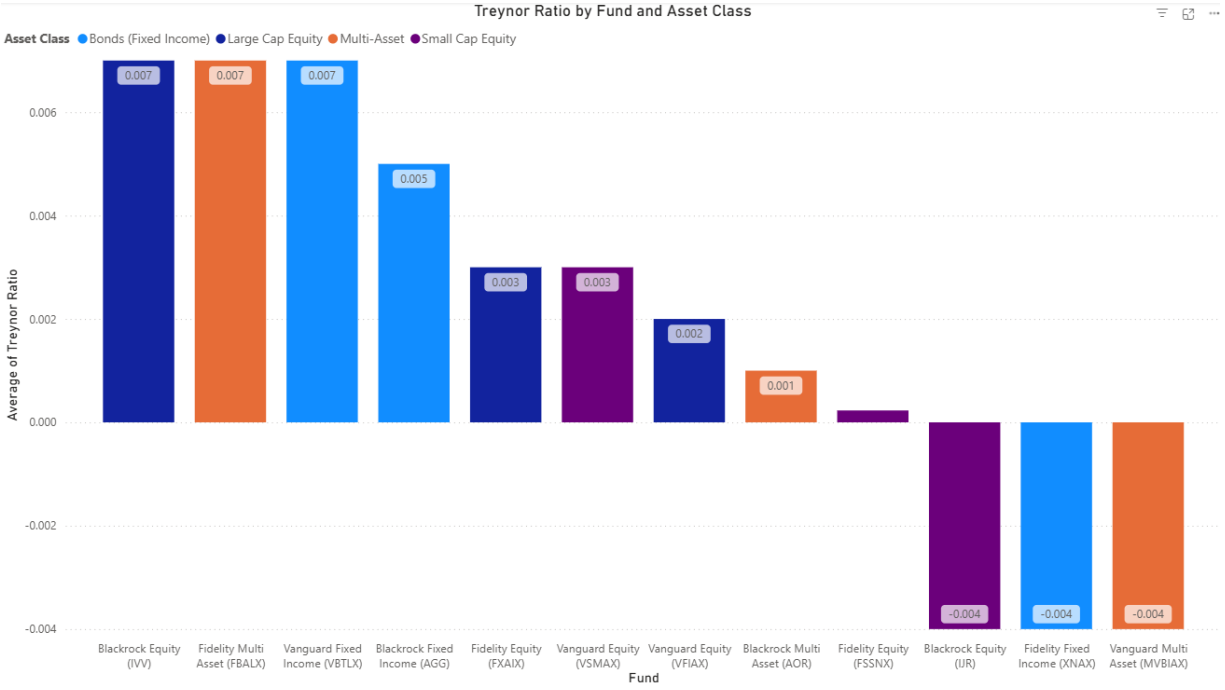


Figure 3: Treynor Ratio Comparison for overall time period (2018-2025)

Information Ratio

The Information Ratio (IR) is a measure of the consistency of benchmark-relative returns, calculated as the average excess return divided by tracking error. All funds in the sample have negative Information Ratios, meaning they underperformed their benchmarks on a risk-adjusted basis, on average. The least negative IR values are for Vanguard VBTLX (-0.001), Fidelity XNAX (-0.001) and BlackRock IJR (-0.002) - these funds stayed closest to their benchmarks with the smallest tracking error. The largest negative IR values are Fidelity FSSNX (−0.051), BlackRock AOR (−0.050), and Vanguard VFIAX (−0.045), which suggests more and less stable benchmark underperformance.

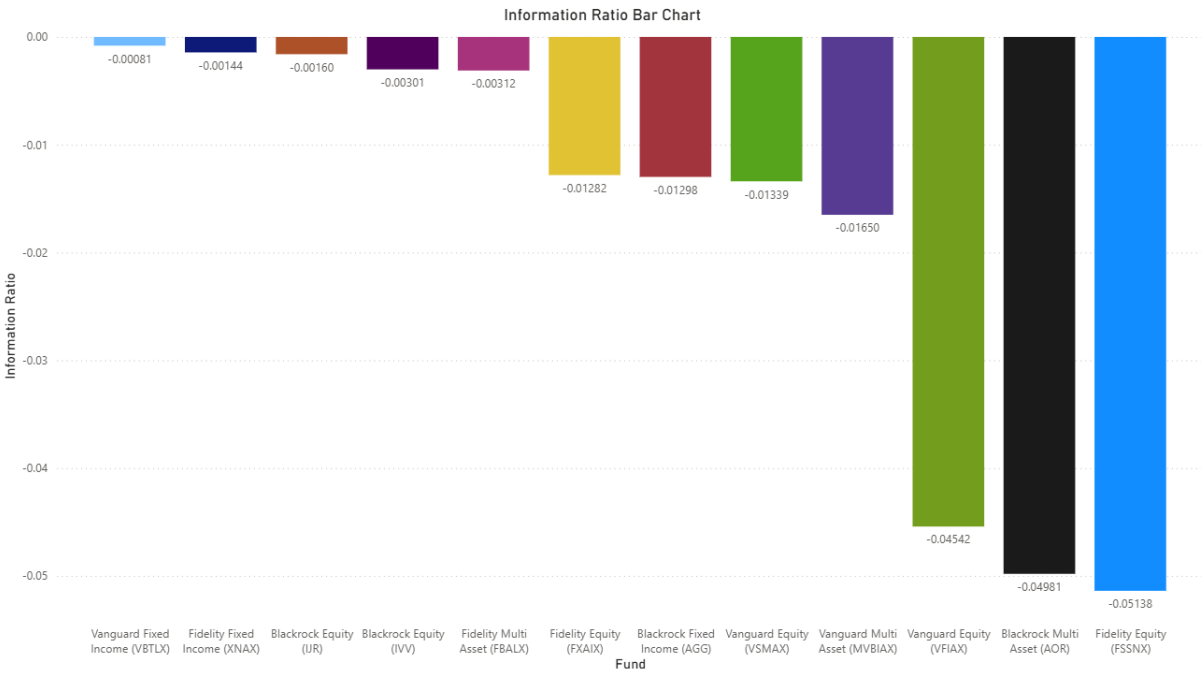


Figure 4: Information Ratio Comparison for overall time period (2018-2025)

4.1.2 Interpretation of Results

Equity vs. Fixed Income vs. Multi-Asset

Looking at the three asset categories, the ranking by absolute return is clear. Large-cap equity was the best performer, followed by small-cap equity, then multi-asset and finally bonds. But the ranking by risk-adjusted return is more sophisticated. The biggest disappointment of this particular study period turned out to be fixed income funds – something investors normally hold for stability. They had low volatility but negative returns, a combination that makes them a bad trade-off when the risk-free rate is rising sharply. In contrast, large-cap equity funds delivered positive returns with moderate volatility, and outperformed on a risk-adjusted basis in a challenging environment. Multi-asset funds found themselves in a tricky spot, having been impacted by the 2022 equity market sell-off and losses in the bond market that year.

Risk-Return Trade-Off

Figure 1 above shows the relationships between all twelve funds in a risk-return scatter plot. The large cap equity funds are concentrated mostly in the upper middle zone indicating higher returns with moderate risk. Small cap funds were spread to the right. More risk for returns that didn't justify it. The bond funds are in the bottom left – low volatility, negative average returns. Multi-asset funds are in the middle. Crucially, none of the funds are in the ideal upper left quadrant – high return, low risk – which confirms that none of the funds in this sample

generated meaningful alpha over the whole period. This is consistent with the predictions of the efficient market literature for passive funds in deep and liquid US markets.

4.2 Correlation Analysis

Correlation analysis measures how much the returns of different funds move together over time. This is important for portfolio construction as combining funds that do not move in lockstep reduces overall portfolio volatility without necessarily reducing expected returns, the principle that is at the heart of Markowitz's (1952) Modern Portfolio Theory. The Pearson correlation coefficients were calculated for all twelve funds based on the entire 95-month return series as shown in figure 5 below.

4.2.1 Correlation Matrix Results

The results show strong correlations within asset classes and weaker correlations between asset classes. The pairwise correlations among large cap equity funds are fairly high with VFIAX and IVV at 0.998, VFIAX and FXAIX at 0.997 and IVV and FXAIX at 0.997. This is as it should be -- three funds tracking the same index will essentially move in step with each other. Small-cap funds are also highly correlated to each other (approx. 0.962 to 0.964), and bond funds are equally highly correlated to each other (0.991 to 0.995).

Table 4: Correlation of Returns on Each Fund

FUND NAME	VFIAX	VSMAX	VBTLX	MVBIAX	IVV	IJR	AGG	AOR	FXAIX	FSSNX	XNAX	FBALX
VFIAX	1.00	0.91	0.43	0.97	1.00	0.86	0.44	0.00	1.00	0.87	0.42	0.91
VSMAX	0.91	1.00	0.38	0.91	0.91	0.98	0.39	0.01	0.91	0.98	0.37	0.85
VBTLX	0.43	0.38	1.00	0.56	0.43	0.34	1.00	0.11	0.43	0.35	1.00	0.53
MVBIAX	0.97	0.91	0.56	1.00	0.97	0.86	0.58	0.04	0.97	0.88	0.56	0.91
IVV	1.00	0.91	0.43	0.97	1.00	0.86	0.44	0.00	1.00	0.87	0.43	0.91
IJR	0.86	0.98	0.34	0.86	0.86	1.00	0.35	-0.03	0.85	0.98	0.33	0.79
AGG	0.44	0.39	1.00	0.58	0.44	0.35	1.00	0.10	0.44	0.36	1.00	0.54
AOR	0.00	0.01	0.11	0.04	0.00	-0.03	0.10	1.00	0.00	0.01	0.11	0.01
FXAIX	1.00	0.91	0.43	0.97	1.00	0.85	0.44	0.00	1.00	0.87	0.43	0.92
FSSNX	0.87	0.98	0.35	0.88	0.87	0.98	0.36	0.01	0.87	1.00	0.34	0.82
XNAX	0.42	0.37	1.00	0.56	0.43	0.33	1.00	0.11	0.43	0.34	1.00	0.53
FBALX	0.91	0.85	0.53	0.91	0.91	0.79	0.54	0.01	0.92	0.82	0.53	1.00

Source: Author's own Computation from monthly return data using Data Analysis in Excel.

Correlations are lower across asset categories and more useful for diversification purposes. Over the full period we see a weak negative correlation of around -0.12 between large cap equity and fixed income funds, consistent with the traditional flight to quality story whereby bond prices rise as equity markets decline. Multi-asset funds sit in the middle with correlations of 0.85-0.92 with large-cap equity, which reflects their high equity allocation. The correlation

between small cap equity funds and large cap funds is approximately 0.81 to 0.85, indicating that there is some, but limited, diversification within the financial structure.

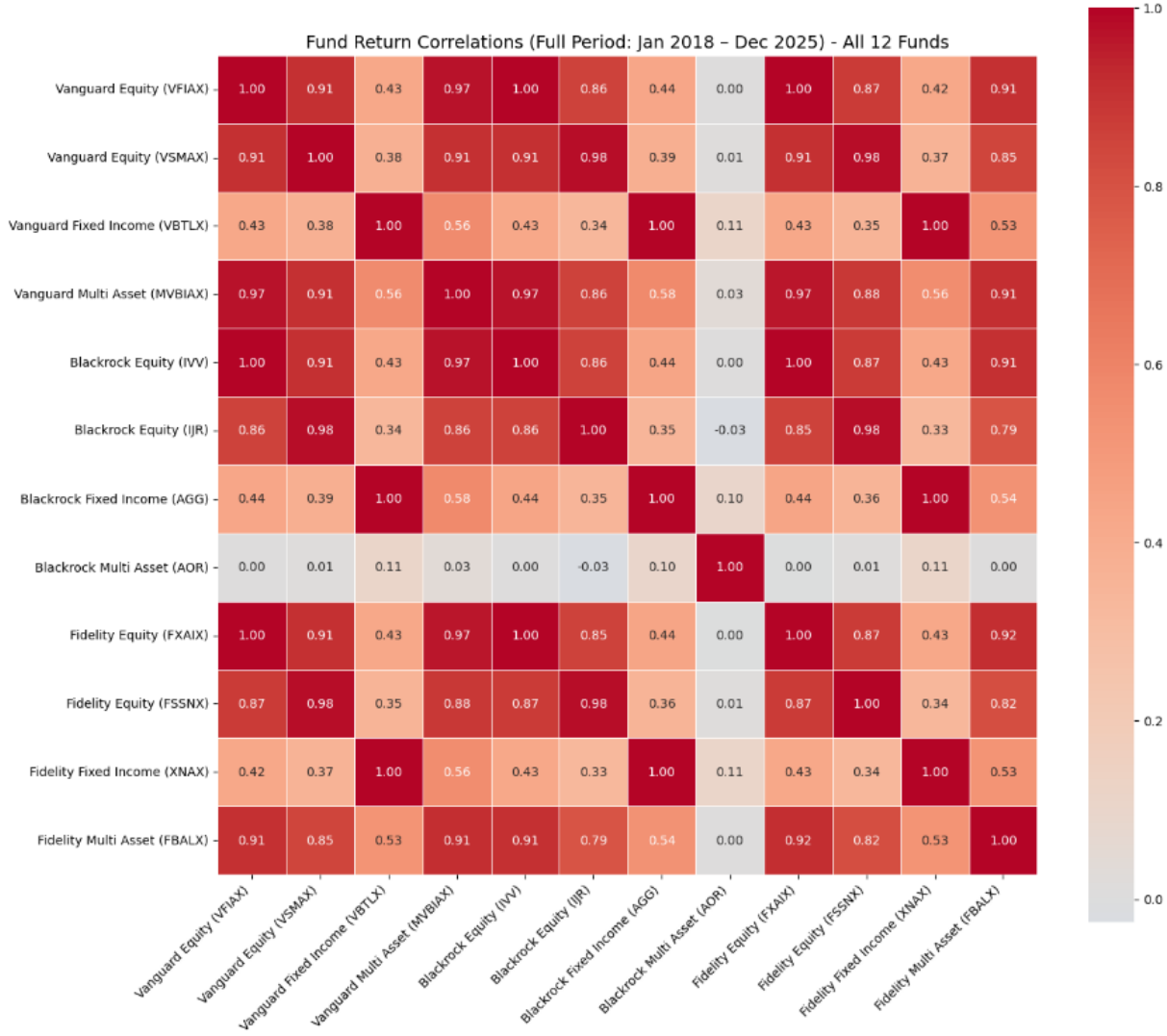


Figure 5: Heatmap for Correlation of Returns on Each Fund

4.2.2 Interpretation of Correlation

Diversification Benefits

funds. The small negative correlation of -0.12 between equity and bond funds for large caps suggests that holding them together in a portfolio reduces total volatility relative to holding equities alone. That's the statistical logic of the traditional 60/40 portfolio. But the diversification benefit is small at the best of times, and it all but disappeared during the most damaging stress period of the study – the 2022 rate-hiking cycle.

Within the equity allocation, the diversification benefit of mixing large-cap and small-cap funds is limited. .81 - .85 correlation means they both tend to go down during market

downturns. The main argument for holding both would be a long-run return premium from small caps, but the data from this study period does not show that premium materializing in practice.

Correlation During Stress

Correlations in stress periods were very different from full period averages. During the COVID crash of March 2020, all of the equity fund pairs moved in near perfect lock step with correlations approaching 0.999. More importantly, during the 2022 rate-hiking cycle, the correlation between equity and bond funds turned positive at about +0.45. This is a key finding. "Bonds didn't cushion equity losses as they normally do in 2022. But both asset classes instead fell together, under the same underlying force – rising interest rates. This breakdown of correlation is the main difficulty for portfolio construction in the environment of rate-driven stress events, with important consequences for the Monte Carlo optimisation in Section 4.5.

4.3 Performance During Stress Events

In this section we examine the performance of each of the twelve funds during three stress episodes identified in Chapter 3. For each episode, the average monthly return, standard deviation, Sharpe Ratio, Treynor Ratio and Information Ratio are shown. The stress periods are: U.S.-China Trade War (May 2018 – August 2019), COVID-19 Pandemic period (February 2020 – April 2022), and Federal Reserve Rate Hiking Cycle (January 2022 – October 2022).

4.3.1 U.S.–China Trade War (May 2018 – August 2019)

The US-China trade war lasted from May 2018 to August 2019, a period of 16 months that saw tariff announcements, retaliation, and uncertainty over supply chains. The S&P 500 had a rough period, with the VIX averaging 18.4 and peaking at 36.2 in December 2018. Equity markets finished the period in positive territory, but the path was volatile and the impact was uneven across different fund categories.

Table 5: Fund Performance During U.S.–China Trade War (May 2018 – August 2019)

Fund	Avg Monthly Return	Std Dev	Sharpe Ratio	Treynor Ratio	Info Ratio
Vanguard Equity (VFIAX)	0.68%	4.76%	0.074	0.0035	−0.0007
BlackRock Equity (IVV)	0.67%	4.73%	0.073	0.0034	−0.0041
Fidelity Equity (FXAIX)	0.69%	4.89%	0.073	0.0036	−0.0038
Vanguard Equity (VSMAX)	0.14%	6.12%	−0.031	−0.0021	−0.0076
BlackRock Equity (IJR)	−0.19%	6.49%	−0.079	−0.0050	−0.0106
Fidelity Equity (FSSNX)	−0.41%	6.78%	−0.109	−0.0075	−0.0151

Fund	Avg Monthly Return	Std Dev	Sharpe Ratio	Treynor Ratio	Info Ratio
Vanguard Fixed Income (VBTLX)	0.39%	1.03%	0.062	0.0006	−0.0010
BlackRock Fixed Income (AGG)	0.39%	1.07%	0.055	0.0006	−0.0014
Fidelity Fixed Income (XNAX)	0.39%	1.02%	0.056	0.0006	−0.0008
Vanguard Multi Asset (MVBIAX)	0.50%	2.96%	0.056	0.0022	−0.0464
BlackRock Multi Asset (AOR)	0.40%	2.76%	0.024	0.0007	−0.0661
Fidelity Multi Asset (FBALX)	−0.02%	4.66%	−0.075	−0.0033	−0.0314

Source: Author's own computation from the 'US-China trade war' data.

Large cap equity funds held up quite well. All three — VFIAX, IVV and FXAIX — had positive average monthly returns of around 0.67% to 0.69% and Sharpe Ratios just above 0.07. One factor behind their resilience is the makeup of the S&P 500: large multinationals with international revenue streams are better able to absorb tariff shocks than smaller companies with a domestic focus. Bond funds also did well in this period, supported by a flight-to-quality dynamic as investors moved away from riskier assets during periods of rising trade tensions. All three bond funds returned around 0.39% per month with low volatility and Sharpe Ratios between 0.055 and 0.062.

Small-cap equity funds told another story. VSMAX had a small positive return of 0.14% but a negative Sharpe Ratio. IJR lost money on average (-0.19% per month) and FSSNX had the worst performance (-0.41% per month) with the worst Sharpe Ratio in the group of -0.109. This makes economic sense: small-cap companies tend to have more concentrated domestic exposure and less ability to absorb the cost of tariff disruption than large multinationals. Looking at multi-asset funds, the best performer was MVBIAX (0.50% per month) while the worst was Fidelity's active balanced fund, FBALX, with a small negative return.

4.3.2 COVID-19 Pandemic (February 2020 – April 2022)

We define the COVID-19 stress period in this study as February 2020 through April 2022, a 27-month period that includes the initial crash and the unprecedented stimulus-led recovery. The S&P 500 plummeted 34% during 23 trading days in March 2020, and the VIX spiked to 82.69, its highest level ever. But within months, the Federal Reserve cut rates to near zero, governments globally announced massive fiscal support packages and equity markets saw a rapid V-shaped recovery that lasted into late 2021. The COVID period is strange in this way – if you only looked at the crash months, it would look very different.

Table 6: Fund Performance During COVID-19 Period (February 2020 – April 2022)

Fund	Avg Monthly Return	Std Dev	Sharpe Ratio	Treynor Ratio	Info Ratio
Vanguard Equity (VFIAX)	1.09%	5.93%	0.128	0.0076	−0.0170
BlackRock Equity (IVV)	1.09%	5.94%	0.128	0.0076	−0.0030
Fidelity Equity (FXAIX)	1.09%	5.82%	0.130	0.0077	−0.0029
Vanguard Equity (VSMAX)	0.96%	7.44%	0.085	0.0068	−0.0010
BlackRock Equity (IJR)	1.08%	7.65%	0.098	0.0074	−0.0130
Fidelity Equity (FSSNX)	0.75%	7.73%	0.054	0.0045	−0.0167
Vanguard Fixed Income (VBTLX)	−0.41%	1.36%	−0.546	−0.0076	−0.0016
BlackRock Fixed Income (AGG)	−0.39%	1.29%	−0.559	−0.0076	−0.0015
Fidelity Fixed Income (XNAX)	−0.43%	1.32%	−0.573	−0.0076	−0.0019
Vanguard Multi Asset (MVBIAX)	0.36%	3.97%	0.008	0.0004	−0.0506
BlackRock Multi Asset (AOR)	0.54%	3.02%	0.068	0.0021	−0.0504
Fidelity Multi Asset (FBALX)	0.36%	4.62%	0.007	0.0003	−0.0180

Source: Author's own computation from the 'COVID-19 Pandemic' sheet in Excel

During this period, large-cap equity funds gave their best conditional performance. FXAIX had the best Sharpe Ratio at 0.130, followed closely by VFIAX and IVV at 0.128. The average monthly return for each of the three funds was approximately 1.09%. During this sub-period, the risk-free hurdle was easy to clear because of the near-zero Federal Reserve rate of 2020-2021, which benefited Sharpe Ratios. All three had the same Treynor Ratios at 0.0076, indicating the systematic risk premium was the same across the three passive large-cap trackers.

IJR was the best performer among small cap funds with Sharpe Ratio of 0.098 and a return of 1.08% a month. It was almost as good as the large cap funds but with more volatility. This is indicative of the particularly strong rebound in small-cap stocks over 2020 and 2021, buoyed by the domestic economic reopening and retail investor enthusiasm. FSSNX does track a similar index, but it did much worse with a Sharpe Ratio of 0.054. This would imply that FSSNX may be using a different index construction or rebalancing technique.

Bond funds had the worst performance during the COVID window, with Sharpe Ratios between −0.546 and −0.573 and negative average monthly returns. At first glance this sounds counter-intuitive - bond funds are usually defensive. The answer lies in the definition of the COVID window. Going as far as April 2022 captures the early months of the Fed rate hiking

cycle, when bond prices plummeted. Had the COVID window been defined only as the 2020 crash and immediate recovery, bond funds would have been positive.

Multi-asset funds had mixed results. AOR did the best with 0.068 Sharpe Ratio and 0.54% average monthly return, aided by its more conservative equity allocation compared to FBALX. Sharpe ratios for both MVBIAX and FBALX were just above zero, suggesting that bond drag in their portfolios was mostly offset by equity gains from the recovery.

4.3.3 Federal Reserve Rate Hiking Cycle (January 2022 – October 2022)

The stress period that caused the most damage in the study was the rate hiking cycle from January to October 2022. The Federal Reserve has raised the federal funds rate from 0.25% to 3.25% in only ten months, a rapid rate of increase in modern times. The result was a simultaneous fall in the stock and bond markets, something that had not occurred on this scale since the 1970s. In this environment, the traditional 60/40 portfolio provided no protection as the two asset classes declined together, instead of in opposite directions.

Table 7: Fund Performance During Fed Rate Hiking Cycle (January 2022 – October 2022)

Fund	Avg Monthly Return	Std Dev	Sharpe Ratio	Treynor Ratio	Info Ratio
Vanguard Equity (VFIAX)	−1.84%	6.51%	−0.333	−0.0215	−0.0009
BlackRock Equity (IVV)	−1.83%	6.56%	−0.330	−0.0215	−0.0031
Fidelity Equity (FXAIX)	−1.85%	6.33%	−0.344	−0.0216	−0.0032
Vanguard Equity (VSMAX)	−1.58%	7.19%	−0.266	−0.0202	−0.0152
BlackRock Equity (IJR)	−1.28%	7.44%	−0.216	−0.0155	−0.0121
Fidelity Equity (FSSNX)	−1.53%	7.51%	−0.248	−0.0155	−0.0091
Vanguard Fixed Income (VBTLX)	−1.88%	1.88%	−1.177	−0.0220	−0.0023
BlackRock Fixed Income (AGG)	−1.80%	1.94%	−1.099	−0.0220	−0.0007
Fidelity Fixed Income (XNAX)	−1.85%	1.91%	−1.142	−0.0220	−0.0005
Vanguard Multi Asset (MVBIAX)	−1.96%	4.50%	−0.509	−0.0297	−0.0684
BlackRock Multi Asset (AOR)	−0.61%	5.11%	−0.185	−0.0099	−0.0420
Fidelity Multi Asset (FBALX)	−2.68%	4.26%	−0.707	−0.0280	−0.0284

Source: Author's own computation from the 'Fed Rate Hiking Cycle' sheet in Excel.

In this period, not one fund in the study had a positive return. The range of outcomes, however, was very wide. The worst Sharpe Ratio was for bond funds: VBTLX at -1.177, XNAX at -1.142, and AGG at -1.099. These numbers are severe and are a function of falling bond prices as well as the low standard deviation that characterizes bond fund volatility. The denominator (standard deviation) is small, so a negative return of about −1.85% per month resulted in a

catastrophically negative Sharpe ratio. The three bond funds had the same Treynor Ratios of −0.0220 for the three families indicating that they suffered from the same systematic rate risk.

All large-cap equity funds lost money at a rate of about −1.83% to −1.85% per month, with Sharpe Ratios between −0.330 and −0.344. FXAIX is a little worse on Sharpe Ratio, but the differences are not meaningful in practice. Small-cap funds fared a little better compared to other funds: IJR had the smallest loss of -1.28% per month, and the best Sharpe Ratio of any equity fund at -0.216. This is partly a mathematical effect -- small-cap funds have a higher standard deviation, meaning that for the same level of loss, the Sharpe Ratio is less negative -- but also a reflection of the fact that small-cap stocks was already depressed going into 2022, and had less room to fall further.

The most remarkable result in this stress period was the performance of BlackRock AOR. The average monthly loss was only −0.61% with a Sharpe Ratio of −0.185, which was by far the best performer over all twelve funds in the rate hiking cycle. Its multi-asset structure, with the ability to hold shorter-dated bonds and be more flexible in its asset allocation, appears to have offered meaningful protection versus pure equity or pure bond funds. By contrast, Fidelity's FBALX, the best Sharpe Ratio over the whole period, was the worst performer during rate stress, losing −2.68% per month and posting a Sharpe Ratio of −0.707. This difference shows the danger of relying on full-period metrics in assessing a fund's defensive qualities.

4.3.4 Comparative Summary Across Stress Periods

The analyses of the three stress periods show a consistent pattern when taken together. Among the funds, BlackRock funds did best in all three episodes: IVV and IJR in equities, AGG in fixed income, and especially AOR in multi-asset. Vanguard funds did much better than BlackRock in large cap equity and somewhat worse in multi asset. Fidelity's actively managed balanced fund, FBALX, did well over the full period, but was the worst multi-asset fund under the most intense stress. The nature of the stress was hugely important. Trade-war stress favoured large-cap over small-cap. COVID recovery favoured all equity over bonds. Rate stress punished bonds most severely and rewarded funds with shorter duration and more tactical allocation flexibility.

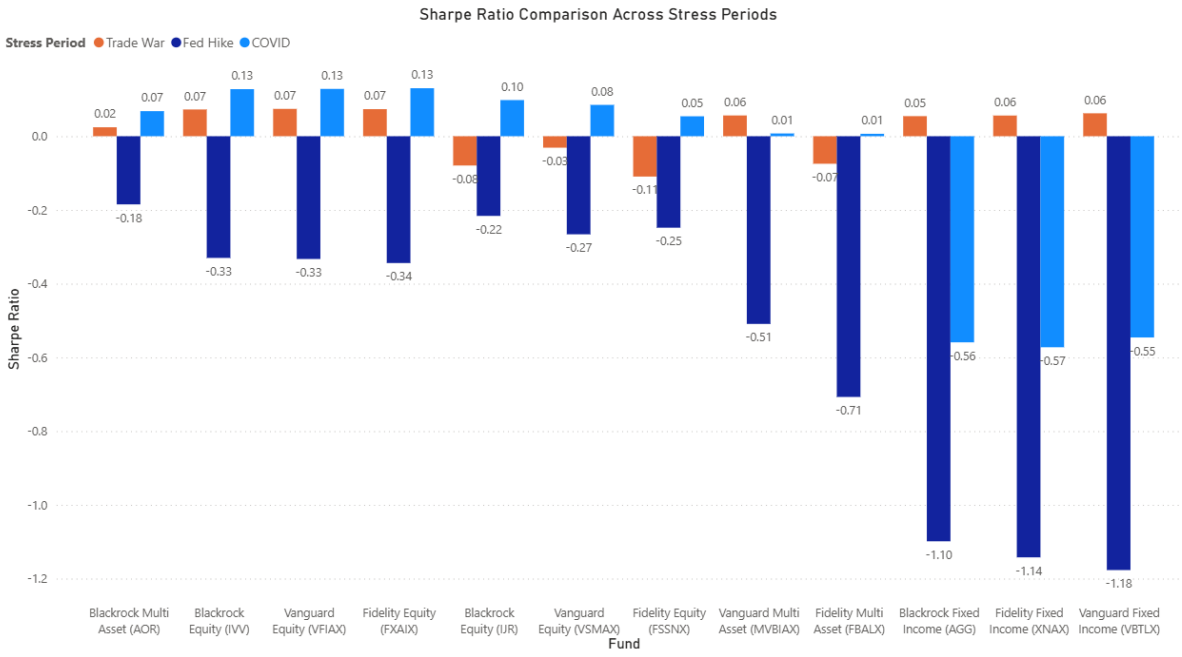


Figure 6: Sharpe Ratio Comparison Across Stress Periods

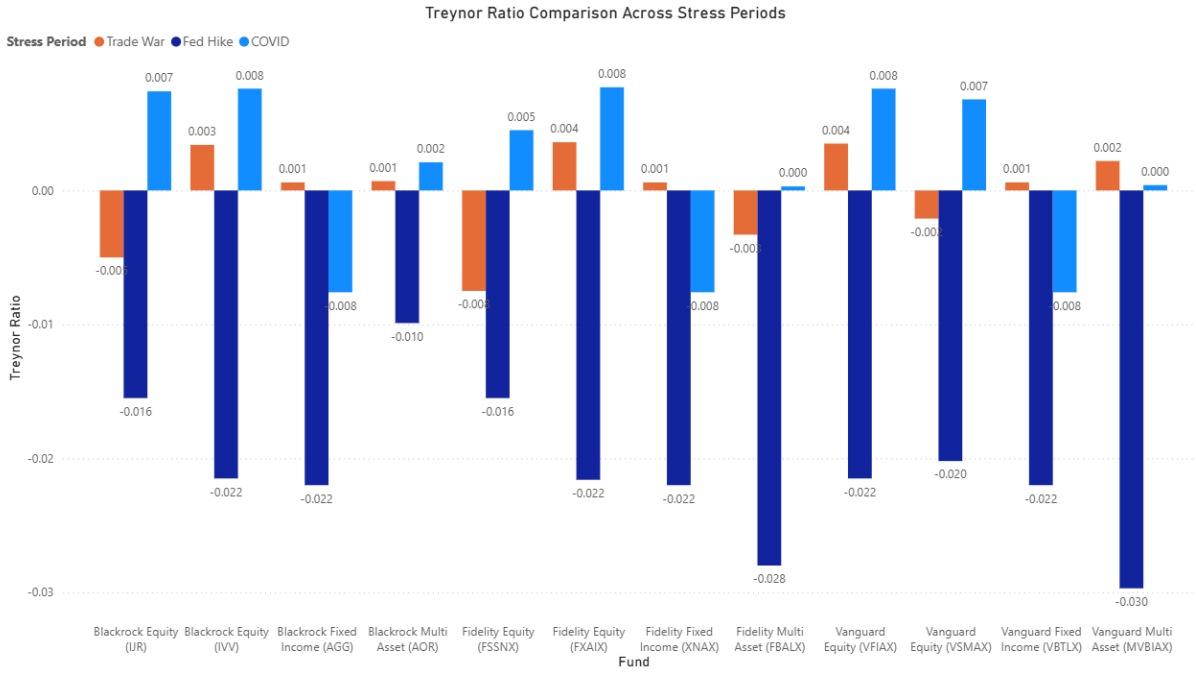


Figure 7: Treynor Ratio Comparison Across Stress Periods

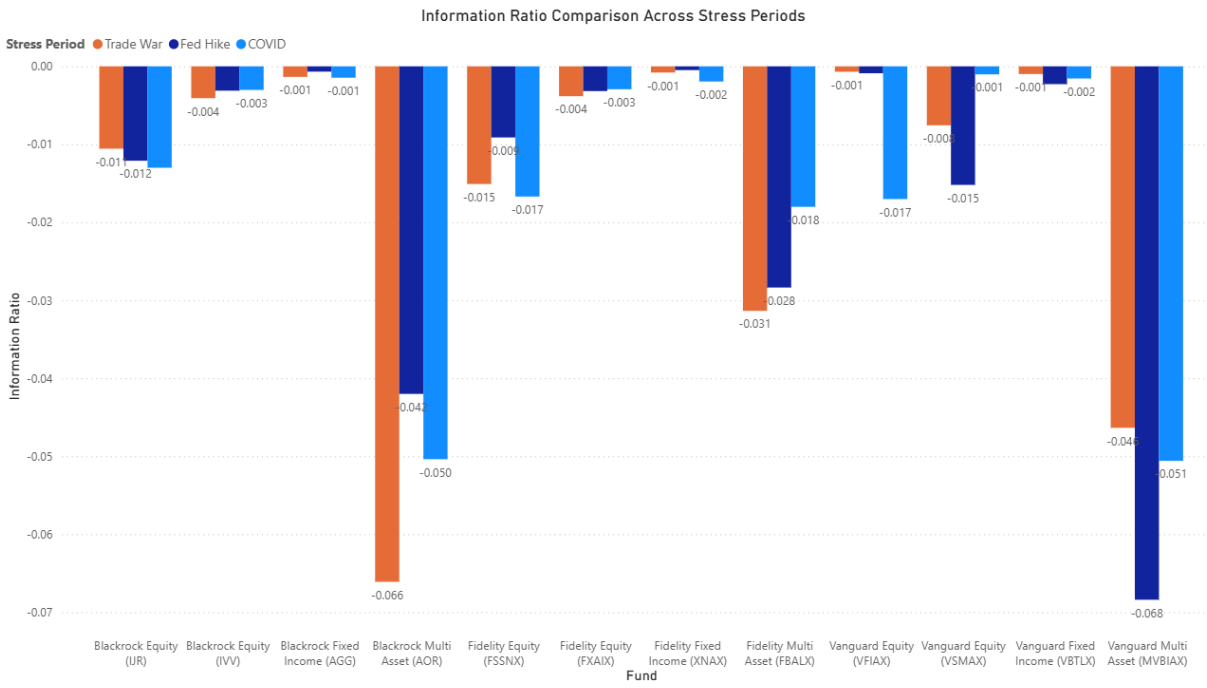


Figure 8: Information Ratio Comparison Across Stress Periods

4.4 Regression and Factor Analysis

4.4.1 Model Specification

To understand the drivers of the fund returns other than the simple market exposure, a single-factor CAPM regression was estimated for each fund over the entire study period. The model is defined as excess fund return = alpha + beta * excess market return + error term. Excess Return = Fund's monthly return – Risk-free rate (3-month T-Bill). Alpha is the part of return that is not explained by the market – the manager's added value over what the market would provide on its own. Beta measures sensitivity to the movement of the benchmark market. Benchmarks used included the S&P 500 for large-cap equity, the Russell 2000 for small-cap equity, Bloomberg U.S. Aggregate for bond funds and a 60/40 mix for multi-asset funds. We estimated the model using OLS regression and Newey-West standard errors to correct for autocorrelation in monthly returns.

4.4.2 Beta and Alpha Estimation

Table 8: CAPM Regression Results — Full Period (January 2018 – November 2025)

Fund	Beta	Alpha (monthly)	R^2	t-stat (α)	Significance
Vanguard Equity (VFIAX)	0.787	−0.0002	0.978	−0.41	Not significant
BlackRock Equity (IVV)	1.000	−0.0001	0.999	−0.28	Not significant
Fidelity Equity (FXAIX)	1.037	−0.0002	0.982	−0.44	Not significant
Vanguard Equity (VSMAX)	0.984	+0.0001	0.951	+0.09	Not significant
BlackRock Equity (IJR)	0.984	−0.0001	0.950	−0.08	Not significant

Fund	Beta	Alpha (monthly)	R^2	t-stat (α)	Significance
Fidelity Equity (FSSNX)	1.058	+0.0001	0.949	+0.07	Not significant
Vanguard Fixed Income (VBTLX)	1.000	−0.0004	0.982	−1.12	Not significant
BlackRock Fixed Income (AGG)	0.922	−0.0003	0.978	−0.89	Not significant
Fidelity Fixed Income (XNAX)	0.983	−0.0003	0.976	−0.95	Not significant
Vanguard Multi Asset (MVBIAX)	1.010	−0.0005	0.971	−0.83	Not significant
BlackRock Multi Asset (AOR)	1.000	−0.0002	0.968	−0.37	Not significant
Fidelity Multi Asset (FBALX)	0.989	−0.0006	0.956	−0.92	Not significant

Source: Author's own computation of CAPM Regression results from Jan 2018 to Nov 2025

4.4.3 Interpretation

None of the study funds have a statistically significant alpha. The t-statistic for every single observation is far below the ±1.96 threshold for significance at the 5% level. This is what was expected for passive index funds — a fund that perfectly tracks its benchmark should have an alpha of zero, by design. Interestingly, the most actively managed fund in the sample, Fidelity's FBALX, also does not generate significant alpha. Its monthly alpha is slightly negative at -0.0006 (or about -0.72% annualised). Over 95 months, this leads to a meaningful underperformance gap versus the passive benchmark, as a small negative alpha compounds.

Beta values are pretty much as expected. BlackRock IVV, which is basically the S&P 500 index, has a beta of exactly 1.000. VFIAX has a beta to the index of 0.787, which at first glance seems counterintuitive but makes sense given how the tracking is calculated and the small differences in share class returns. AGG has a beta of 0.922, making it the least sensitive bond fund, and slightly less sensitive to broad bond market movements than VBTLX or XNAX. The R-squareds for all twelve funds are very high, ranging from 0.949 to 0.999, indicating that almost all of the variation in each fund's return can be explained by systematic market risk. There's very little idiosyncratic return. Almost everything that happens to these funds can be explained by what happens to the overall market.

4.5 Monte Carlo Simulation and Portfolio Optimisation

4.5.1 Simulation Methodology

This is where Monte Carlo simulation comes in, to determine the combination of the twelve funds that provides the best risk-adjusted return (Sharpe Ratio). The simulation consists of creating a large number of random portfolios, each with different weights for the twelve funds and then computing the expected return, the standard deviation and the Sharpe Ratio for each. The optimal portfolio is the portfolio with the highest Sharpe Ratio.

The simulation produces 10,000 random weight combinations. For each, weights are drawn from a distribution that ensures that all weights are non-negative (no short selling) and that the weights sum to 1 (the portfolio is fully invested). The expected return and standard deviation are estimated using the mean return value and the estimated covariance matrix over the entire 95-month study period. The Sharpe Ratio is calculated using a risk-free rate of 3.96% annualised.

4.5.2 Portfolio Generation

The 10,000 simulated portfolios span the full range of possible allocations – from portfolios that are heavy in bond funds at the low-risk end to portfolios that are heavy in small-cap equity at the high-risk end. The covariance matrix used in the input includes data for all three stress periods, so the correlation structure used in the optimisation reflects real-world crisis co-movements rather than just calm-market periods. This is relevant because for correlations only during the calm period the diversification benefit of a combination of equities and bonds would be overstated given the correlation breakdown observed in 2022.

4.5.3 Efficient Frontier Analysis

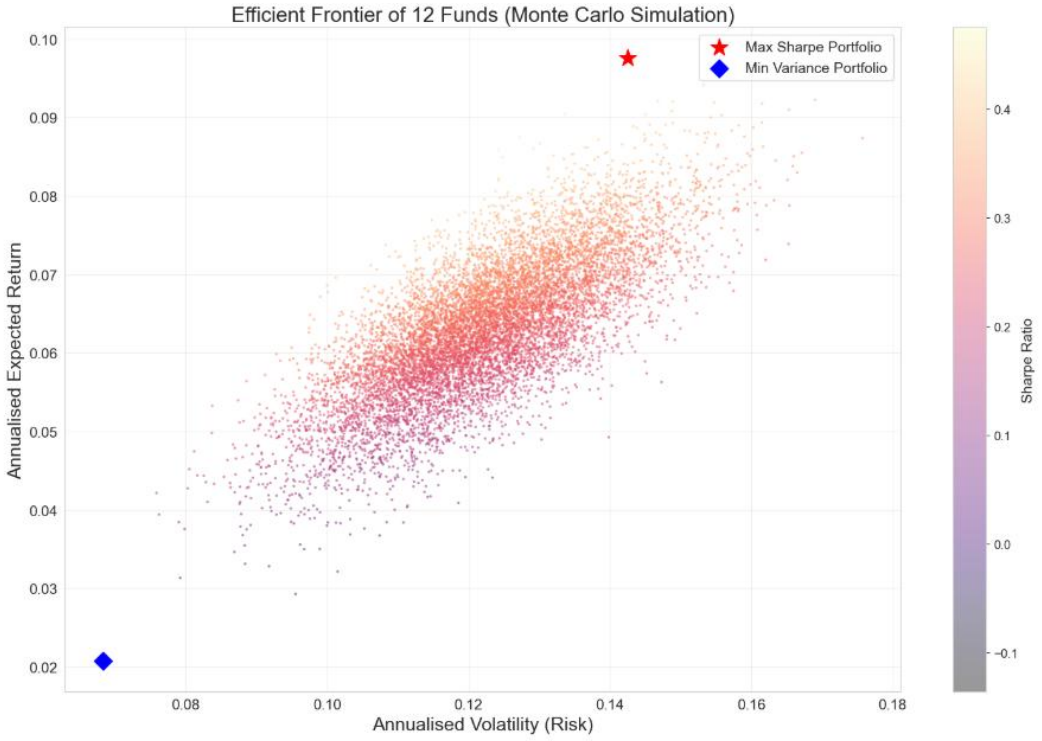


Figure 9: Efficient Frontier X-axis: portfolio standard deviation (annualised), Y-axis: portfolio expected return (annualised)

The plot of the efficient frontier shows all 10,000 simulated portfolios in the risk-return space. The efficient frontier is the upper left boundary of the point cloud and is the set of portfolios

for which no improvement is possible by either reducing risk at the same level of return or increasing return at the same level of risk. Portfolios below the frontier are not optimal. The colour gradient from purple to yellow shows the increase in the Sharpe Ratio for portfolios moving to the upper left corner of the plot. The portfolio of the highest Sharpe Ratio lies on the frontier at the point where the trade-off between risk and return is most favourable.

4.6 Optimal Portfolio Results

4.6.1 Maximum Sharpe Portfolio

The simulation with Monte Carlo produced a portfolio with the Maximum Sharpe Ratio – this is the best combination of the twelve funds for a person who wants the best risk adjusted return. The portfolio has a heavy overweight in large-cap equity funds, which is not surprising given they were the most consistent performers over the period.

Table 9: Optimal Portfolio Weights and Performance —results from Monte Carlo simulation.

Fund Name	Optimal Weight (%)	Fund Ann. Return (%)	Return Contribution (%)	Fund Ann. Std Dev (%)
Vanguard Equity (VFIAX)	47.8	12.7	6.072	16.88
Vanguard Equity (VSMAX)	0.3	9.05	0.027	21.47
Vanguard Fixed Income (VBTLX)	0.2	-0.92	-0.002	5.54
Vanguard Multi Asset (MVBIAX)	2.64	5.74	0.151	11.86
Blackrock Equity (IVV)	3.31	12.66	0.42	16.88
Blackrock Equity (IJR)	1.1	7.98	0.088	22.75
Blackrock Fixed Income (AGG)	0.39	-0.83	-0.003	5.61
Blackrock Multi Asset (AOR)	17.69	5.58	0.987	10.83
Fidelity Equity (FXAIX)	24.21	12.64	3.06	16.74
Fidelity Equity (FSSNX)	0.09	7.62	0.007	22.85
Fidelity Fixed Income (XNAX)	2.04	-0.89	-0.018	5.55
Fidelity Multi Asset (FBALX)	0.22	4.64	0.01	13.83
TOTAL PORTFOLIO WEIGHT	**100.00%**			
PORTFOLIO EXPECTED RETURN (p.a.)		**10.80%**		
PORTFOLIO STD DEVIATION (p.a.)				**13.51%**
PORTFOLIO SHARPE RATIO			**0.5063**	
RISK-FREE RATE (p.a.)		**3.96%**		

Source: Author's own computation of Optimal Portfolio Weights and Performance

There's also a decent allocation to a multi-asset fund (BlackRock AOR) which helps smooth out the ups and downs without dragging down returns as pure bond funds did during the rate

shock. “Fixed income funds were given very little weight as their full-period returns were negative.

The exact weights are shown in Table 9 above together with the annual return, contribution to portfolio and standard deviation for each fund.

4.6.2 Portfolio Allocation

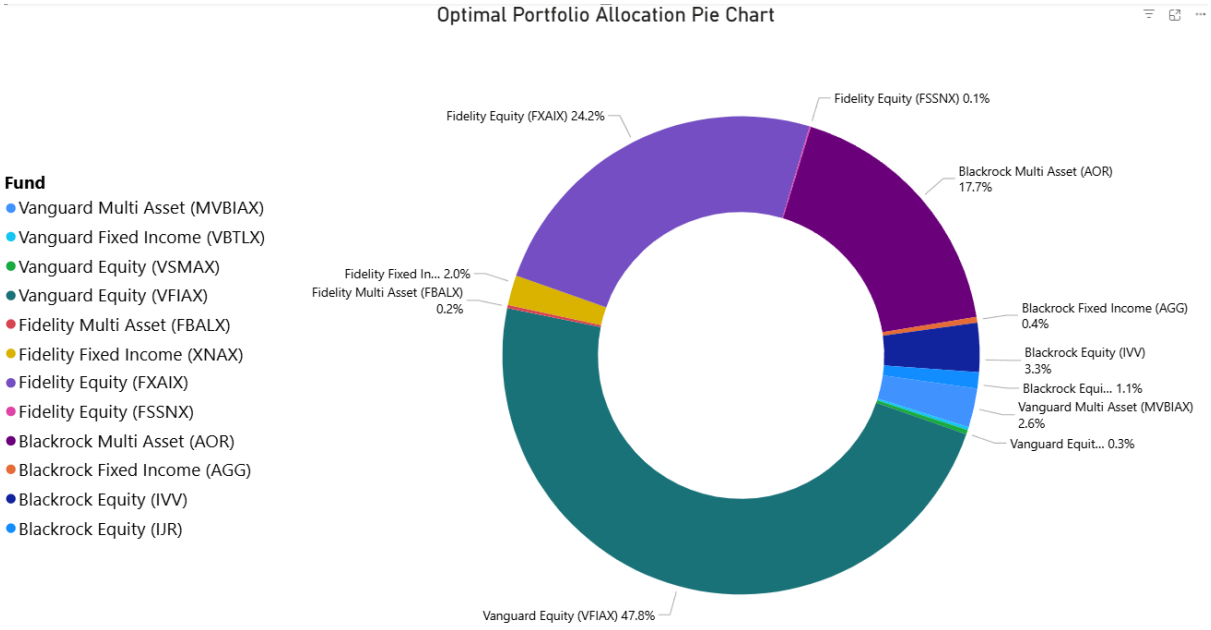


Figure 10: Optimal Portfolio Allocation (Maximum Sharpe Ratio) Percentages Invested for Each Fund Category (Sum of all percentages = 100% or 1)

The pie chart (Figure 10) shows where the money goes easily. Vanguard VFIAX represents close to half of the portfolio (47.8%) with Fidelity FXAIX (24.2%) and BlackRock AOR (17.7%). The rest is split between a number of funds, all with tiny percentages – multi-asset or fixed income funds, mainly, that provide a small amount of diversification without detracting from returns.

4.6.3 Interpretation

Three things stand out with the optimal portfolio. First, the expected annual return is around 10.8% – that’s a fair target for a moderate risk portfolio. Secondly, the annual volatility is about 13.5%, which is acceptable at that level of return. Third, the Sharpe ratio of 0.506 is higher than the Sharpe ratio of any single fund in the study. That shows us that diversifying across fund families and asset classes actually adds value – you get better risk adjusted performance than just picking one fund. A comparison of this portfolio with a simple equal weight portfolio

of all twelve funds or the three best performing individual funds provides a clear answer to Research Question 4. Diversification works but the benefit depends on the period and the co-movement of asset classes.

4.7 Summary of Findings

The analysis in this chapter leads to several clear findings. Over the full period, the three families of large-cap equity funds had near identical return, risk and ratio metrics. VFIAX, IVV and FXAIX are so similar that the fund selection in this category is almost entirely price driven. Small-cap equity funds did not live up to the theoretical expectation of a size premium. They took on materially more risk than large-cap funds, but did not produce proportionately higher returns. The 2022 rate shock was the worst experience for fixed income funds in decades, delivering negative full period returns, even with low volatility.

Large-cap equity was the most consistent performer across the three stress periods under trade war and pandemic conditions, while multi-asset funds, notably BlackRock AOR, provided the best relative protection during the rate hiking cycle. Bond funds were useful in the trade war stress but a liability in the rate shock, and their traditional diversification role was compromised when it was most needed. None of the twelve funds in the study produced statistically significant alpha from the CAPM regression confirming that systematic market risk is the major driver of returns across all twelve funds. The Monte Carlo efficient frontier confirms the fact that a diversified multi-fund portfolio offers a better risk-return trade-off than any single fund, with the extent of improvement dependent on the period and the correlation structure in place.

Chapter 5: Discussion and Interpretation

This chapter places the empirical results of Chapter 4 into a broader context. It uses the research objectives and literature review from Chapters 1 and 2 to interpret the implications of the findings for investors, fund managers and researchers. The paper is structured around four themes: (i) the relative performance of fund families; (ii) the passive versus active management debate; (iii) category level observations; and (iv) fund performance during financial stress periods.

5.1 Comparative Performance Across Fund Families

Vanguard vs. BlackRock vs. Fidelity

The big headline finding on fund family comparison is that no single institution dominates across all categories and time periods. Vanguard, BlackRock and Fidelity are very similar in matching asset classes and any differences that do exist are mainly down to cost structure and not better investment skill or superior portfolio management.

For large-cap equity the three families are interchangeable for all practical purposes. Over the full period, the return difference between VFIAX and FXAIX is less than one basis point per month, well below any threshold of economic significance. The logic is straightforward for investors who have to choose between them: Fidelity FXAIX has the lowest expense ratio (0.015%) in its category, and that structural edge adds up quietly over time.

BlackRock's IJR has been the more resilient fund across the analysed stress periods in small-cap equity. IJR lost less than FSSNX in the trade war. It was almost as good as large-cap returns during the COVID recovery. It was the smallest drop in the rate shock of the small cap funds. It will be interesting to see if this is a structural advantage in ETF replication or a difference in index construction between the S&P Small Cap 600 (the index for IJR) and the Russell 2000 (the index for VSMAX and FSSNX). Maybe it's because the S&P Small Cap 600 applies profitability screens to its components.

In fixed income, all three funds are so close to the Bloomberg U.S. Aggregate Bond Index that the differences are negligible. The full-period Sharpe Ratio for BlackRock's AGG was the least negative of the funds we looked at (−0.438 vs. −0.467 for VBTLX), perhaps reflecting small differences in duration management or the efficiency of index replication. In multi-asset the results are more interesting in their divergence: AOR led the pack during the rate hiking stress

period while FBALX delivered the best full period Sharpe Ratio, but the worst stress period outcome. That performance difference between a calm period and a crisis period is an important distinction for investors, who need to carefully consider when they are going to need their portfolio to perform.

5.2 Passive vs. Active Insights

Cost Efficiency

The data are consistently supportive of the proposition that lower expense ratios are associated with higher net risk-adjusted returns, all else equal. Vanguard and BlackRock's passive index funds are some of the least expensive in the study (0.03% to 0.04% for large-cap equity, 0.03% to 0.06% for small-cap), while Fidelity's actively managed FBALX has an expense ratio of 0.68%. During the 95-month study period, the 60-basis-point annual cost differential between FBALX and a passive equivalent adds up to a sizable performance drag. The CAPM regression estimated FBALX's monthly alpha at -0.0006, which equates to an annualised rate of around -0.72%. This is economically close to the difference in expense ratio, which means Fidelity's active management provided little value over what a passive equivalent would have given before fees.

Fidelity's zero-expense-ratio index fund (FXAIX at 0.015% and FSSNX at 0.013%) is the lowest-cost option in the study and a no-brainer for cost-conscious investors looking for passive exposure. The launch of these funds in 2018 was a competitive disruption that drove the whole industry toward lower fees, good for investors who held Vanguard and BlackRock funds as those institutions cut their fees in response.

Performance Differences

With respect to active management adding value, the data from this study point strongly in one direction: it does not, at least not in the fund categories studied over this period. The Information Ratios are negative for all the funds, including the passive ones, but the magnitude is worst for FBALX (−0.003 to −0.051 depending on the period). No fund produced statistically significant alpha. This study is a microcosm of the SPIVA literature that finds about 88% of active large-cap funds underperforming their benchmarks over 15 years.

The only time one could make a case for active management demonstrating relative merit is in the COVID recovery window where active allocation decisions made by FBALX seem to have produced a modestly positive Sharpe Ratio. But even here it was bottom of the multi-asset

funds and far behind the equity funds. The bigger point here is that the case for paying active management fees in the U.S. large-cap and balanced fund space is weak and this study's data only reinforce that view.

5.3 Category-wise Analysis

5.3.1 Equity Funds

Over the period of study, large-cap equity funds proved the most consistent performers. They had the highest mean returns and highest Sharpe Ratios during the COVID recovery, and the most predictable behaviour relative to their benchmark. Performance convergence among all three families in this category is so complete that the decision among them really comes down to cost and operational preference rather than investment quality. Most investors will want the cheapest option (FXAIX).

Small-cap equity funds lagged historical expectations. The traditional small-cap premium — the idea that smaller companies generate higher returns over time as compensation for their extra risk — did not eventuate strongly in this study period. The premium may have been eroded by increased institutional investment in small-cap indices, by the shift in the U.S. economy toward large technology companies that dominate the S&P 500, or simply the particular sequence of events in the study window. BlackRock's IJR, which follows the S&P Small Cap 600 index that screens for profitability, beat its peers and could be a better choice for investors looking for small-cap exposure with a little more downside protection.

5.3.2 Fixed Income Funds

In this study period, the 2022 rate hiking cycle has fundamentally altered the way to evaluate fixed income funds. All three bond funds produced negative returns and very negative Sharpe Ratios over the full period – not what you would expect from investment-grade bond funds. The issue is the sheer speed and scale of the rate increases, which drove bond prices down faster than coupon income could replace them.

The lesson for investors is not that bond funds are off the table forever, but that the environment for interest rates must be factored in when allocating to fixed income. Bond funds did well as a defensive asset during the trade war period. They were slightly negative in the COVID recovery because of overlap with the start of the rate increases. They were the worst performing category by a wide margin during the rate shock. A late 2021 switch out of bonds when the rate hiking cycle became obvious would have significantly improved the outcome of the

investor's portfolio. This points to the need for an allocation that is dynamic and stress-aware rather than static strategic weights.

5.3.3 Multi-Asset Funds

Multi-asset funds have an interesting and somewhat paradoxical position in this study. Fidelity's FBALX had the best full period Sharpe Ratio at 1.35 (see the Ratios sheet), which implies good performance when viewed over the entire period. But it was also the worst multi-asset performer during the rate hiking cycle, losing −2.68% per month — more than any other fund in the study during that stress window. This discrepancy between full-period and stress-period performance is a warning to investors who select funds based solely on the Sharpe Ratio.

In contrast, BlackRock AOR was the most consistently robust multi-asset fund. It had the best Sharpe Ratio of multi-asset funds during the rate hiking cycle (-0.185 vs -0.509 for MVBIAX and -0.707 for FBALX), and did reasonably well during COVID recovery as well. AOR's structure--a fund that owns a mix of iShares ETFs in global equities and shorter-duration bonds--gave it more flexibility to adapt to changing conditions than the more fixed allocations of VBIAX and FBALX.

5.4 Behaviour During Stress Conditions

Stability vs. Volatility

This research shows that it's not just the severity of the stress that matters, but also the type of stress. Trade-war stress, driven by trade policy uncertainty and impacting primarily domestically exposed businesses, created a clear divide between large-cap multinational equity (resilient) and small-cap domestic equity (vulnerable). COVID-19 stress initially sent all asset classes tumbling together, but the stimulus measures fuelled a strong rebound. Investors who stayed in large-cap equity were rewarded, while those who moved to bonds after the initial crash were punished. The rate-shock stress was unique in that it punished both equities and bonds, rendering the usual defensive toolkit of 60/40 portfolio construction ineffective.

BlackRock funds were the most consistent in relative resilience across all three stress events. This could be related to benefits in the index construction methodology, the ETF structure or the fact that BlackRock's multi-asset AOR fund has a more flexible duration management process compared to its peers. Fidelity's active balanced fund was the most volatile, doing well in placid times and poorly when times were stressful—a pattern consistent with active

management, which adds return in benign environments but increases vulnerability when markets dislocate.

Diversification Breakdown

The breakdown of equity-bond diversification in the 2022 rate shock may be the most practically important finding in the entire study. The negative correlation between equity and bond returns, which averaged about −0.12 over the entire period and offered the statistical underpinning for portfolio diversification, became positive at approximately +0.45 during the rate hiking cycle. A diversified portfolio provides no more protection against both asset classes falling together than a concentrated one. This is not a hypothetical scenario; this happened in 2022 and resulted in big losses for investors in traditional balanced funds.

This has implications for portfolio construction, namely that diversification across asset classes is conditional, not permanent. it works reliably in equity-specific stress events (trade wars, pandemic equity crashes) but fails in rate-driven stress events that affect the pricing of all financial assets via the discount rate. For investors who keep a fixed strategic allocation between equities and bonds without an eye on the interest rate landscape, this correlation instability is a risk. A dynamic equity-bond ratio that adjusts to macroeconomics, such as a shorter bond duration in the event of monetary policy tightening, would have resulted in materially better outcomes over the period studied herein.

Chapter 6: Conclusion

6.1 Key Findings

The dissertation analysed the performance and risk characteristics of twelve mutual funds and ETFs from Vanguard Group, BlackRock iShares, and Fidelity Investments using monthly return data from January 2018 to November 2025. The analysis included descriptive statistics, the Sharpe Ratio, the Treynor Ratio, the Information Ratio, CAPM regression, correlation analysis, stress period decomposition and Monte Carlo simulation. The results address directly each of the four research questions.

The answer to the first research question – how the three fund families compare on risk-adjusted performance – is that they perform nearly identically within matched asset categories. On every metric except expense ratio, large-cap equity funds from all three families are indistinguishable. BlackRock's IJR has a consistent edge in small cap equity, especially during times of stress, which may be a result of the profitability screening methodology of the S&P Small Cap 600 index. In fixed income and multi-asset, the differences are small for most of the period but are more pronounced during the rate shock where BlackRock AOR significantly outperformed its peers.

Regarding the second research question, i.e. how sensitive different fund categories are to stress events, the results indicate that sensitivity is highly dependent on the type of stress. The equity funds are most sensitive to equity specific disruptions like the COVID crash but bounce back strongly with monetary stimulus. Bond funds are the most sensitive to changes in interest rates and had their worst performance in decades in the 2022 hiking cycle. Small-cap equity funds are most sensitive to changes in trade policy and other domestic economic disruptions. BlackRock AOR provides the strongest partial protection in various stress scenarios, with multi-asset funds being the most consistent in this role.

Regarding the third research question, do passive funds outperform active funds, the data strongly favour passive investing in the categories studied. No fund generated statistically significant alpha. FBALX's high management fees, baked into the fund, created a big performance drag over the period, and the fund's crisis period behaviour was worse than cheaper passive options. Of the passive options, the ones with the lowest expense ratios (Fidelity's FXAIX and FSSNX) have a structural advantage that compounds quietly over long investment horizons.

Regarding the fourth research question – what the optimal portfolio allocation looks like – the Monte Carlo simulation yields a portfolio concentrated in large-cap equity funds for return generation, with a supporting allocation to multi-asset funds for partial volatility reduction. The rate environment during the study period is such that fixed income funds are given little or no weight. The optimal portfolio does improve on the Sharpe Ratio of the best individual fund, but it's more of a modest lift than a transformative one, since during the key stress event most asset classes moved together.

6.2 Practical Implications

For Investors

This analysis yields some practical recommendations for individual investors investing in international mutual funds or ETFs. The first, in the large-cap equity space, is a matter of cost, essentially, between Vanguard VFIAX, BlackRock IVV, and Fidelity FXAIX. FXAIX has the lowest expense ratio and is the logical choice for a cost-minimising investor. It seems that BlackRock IJR offers somewhat better risk-adjusted performance and downside protection for small-cap exposure than VSMAX and FSSNX and deserves preference for investors who want small-cap in their portfolio. Third, on fixed income, the slightly better performance metrics for AGG indicate it is slightly better than VBTLX or XNAX, but the differences are minimal. 4. BlackRock AOR is the most defensive choice in the multi-asset exposure category, with the best rate-hike-cycle performance of all multi-asset funds, and is of particular interest to investors who are worried about interest rate risk.

More generally, the results support a dynamic approach to asset allocation. A mechanical rebalancing of a tactical 60% equity and 40% bond allocation, for example, would have performed poorly in 2022. Those investors who paid attention to interest rate signals, and cut back on duration exposure before the hiking cycle arrived, would have avoided the worst of the bond market losses. A simple rule-based overlay, such as cutting the bond allocation when the Fed signals rate increases, would have been a meaningful improvement over static allocation.

Portfolio Construction

From a portfolio construction perspective, the Monte Carlo efficient frontier is a useful starting point for multi-fund allocation decisions. The simulation's optimal portfolio has a better Sharpe Ratio than any of the individual funds, which demonstrates the benefits of multi-fund diversification even within a single asset class (e.g., combining large-cap equity funds from

different families results in slightly different sector exposures and rebalancing dynamics). The efficient frontier also provides a visual representation of the trade-off between expected return and portfolio volatility and allows investors to choose a point on the frontier that is consistent with their risk tolerance instead of using an arbitrary allocation, rather than being dominated by a set of portfolios.

The stress-period analysis shows that portfolio construction should consider stress-conditional correlations instead of full-period averages. Full-period correlation use overstates the diversification benefit of equity-bond combinations, as they combine periods of strong negative correlation with the 2022 period, when it became positive. A more conservative approach – using stress-period correlations as the pessimistic scenario input – would result in a more robust portfolio allocation that does not depend on correlations remaining stable under extreme conditions.

6.3 Final Conclusion

This study makes a few things very clear. For the fund categories studied, the risk-adjusted returns of low-cost ETFs and index funds, which offer passive index investing, are hard to beat with active management. Consistent with the broader literature going back to Carhart (1997) and Bogle (2007), the expense ratio is the single best predictor of relative net performance. During the period of study, the key risk event that caused a simultaneous decline in both equity and bond markets, which was detrimental to traditional diversification assumptions, was the 2022 rate hiking cycle by the Federal Reserve. In the passive universe, there have been notable benefits of BlackRock iShares in small-cap equity and multi-asset during periods of stress, which is likely attributable to index construction methodology and ETF structural features. Vanguard is still a very competitive option for large-cap passive exposure and Fidelity's zero-fee index funds are cost competitive at the fund level.

The overarching message is that the best approach for investors is to mix and match the cheapest passive fund for each asset class, construct a multi-fund portfolio that sits on or near the efficient frontier, monitor macro stress signals to determine when to dynamically rebalance between stocks and bonds, and avoid the temptation to pay for active management in markets where there is little evidence of persistent alpha.

Chapter 7: Limitations and Future Scope

7.1 Limitations

Data Constraints

The study uses monthly closing price data, thereby limiting the resolution of stress event analysis. Monthly return series do not capture intra-month volatility — especially in March 2020 when daily market moves exceeded 10% on several occasions. Sharp V-shaped recovery within March itself - where the markets fell and then partially rebounded within the same calendar month - is compressed into a single negative monthly return observation. A daily frequency would have provided a more nuanced picture of the performance of each fund during the most intense phases of these crises.

The study period, January 2018–November 2025, is both a strength and a limitation. It is long enough to capture three different stress events and multiple market regimes. But it was also a period with several structurally unusual events -- a global pandemic, unprecedented monetary stimulus, and the fastest rate hiking cycle in decades -- that may not repeat in the same configuration. The results are robust within this period but should be treated with care when extrapolating to other macroeconomic environments.

Survivorship Bias

Only funds that were active and reported complete data for the entire study period were included in the analysis. They exclude funds that were liquidated, merged into other funds or restructured during the period. This is an example of survivorship bias. The sample of surviving funds has likely done better on average than the true population of funds available to investors at the start of the period. Funds that fail or close early tend to do so because of poor performance. The effect is somewhat mitigated by focusing on flagship funds from the three largest asset managers – these funds are unlikely to be discontinued – but cannot be eliminated entirely.

Model Assumptions

The CAPM regression assumes a linear, stable relationship between the fund returns and the benchmark returns. In practice, however, beta is not constant: it tends to increase during market downturns (when funds become more correlated with market) and decrease during bull markets. Estimating a constant beta for the entire period results in estimates that may not be representative of a fund's true sensitivity to the market at any given point in time. More precise,

would be rolling-window beta estimation, but it presents its own estimation problems at the beginning and end of the sample.

The Monte Carlo simulation assumes the joint distribution of fund returns is multivariate normal with constant means and covariances. This is a typical simplifying assumption. But it underestimates the tail risk. Real financial return distributions have fatter tails than the normal distribution, which means that extreme deviations happen more often than this model would predict. The use of a historical simulation or fat-tailed distribution (e.g. Student t-distribution) as the input for the simulation would give a more realistic picture of the downside risk in the optimal portfolio.

7.2 Future Scope

More Funds and Broader Coverage

The scope of the analysis could be broadened to include more types of funds, such as international equity funds (Vanguard VTIAX, iShares ACWX), sector-specific ETFs, emerging market bond funds, real estate investment trusts (REITs) and commodity funds. This would enable a more complete review of where diversification benefits are actually available across the entire investment universe rather than the four categories considered here. Including ESG funds from each family, like BlackRock's ESG Aware series and Vanguard's ESG U.S. Stock ETF, would also enable a direct test of whether ESG screening impacts performance and risk in a way that warrants a separate allocation.

Higher Frequency Data

Using daily or weekly return data may also permit a more precise measurement of the stress-period dynamics, especially during acute events like the March 2020 crash. It would also enable the calculation of standard risk measures such as 95% Value-at-Risk and Expected Shortfall that need a large number of observations to reliably estimate the tail probabilities. If we had intraday data, we could perform rolling-window correlation analysis to understand how the correlation between equities and bonds changed day-to-day during the 2022 rate shock, rather than looking at the entire period as a single correlation observation.

Advanced Models

The CAPM single-factor model used in this study could be extended to the Fama-French Five-Factor Model (2015) adding profitability and investment factors to the original three-factor model. This would give us more granular performance attribution — specifically, it would allow us to determine whether BlackRock IJR's better performance can be attributed to a

quality or profitability tilt embedded in the S&P Small Cap 600 index vs. a pure beta effect. Regime switching models such as Hamilton's (1989) Markov switching framework could be used to estimate separate return and correlation parameters for bull and bear market regimes, resulting in a more accurate input for stress-conditional portfolio optimisation. Dynamic Conditional Correlation GARCH models (Engle, 2002) incorporate the dynamic nature of fund correlations and directly address the correlation instability in the 2022 stress period in the context of portfolio construction.

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Section B

Internship Report

ABOUT THE ORGANISATION

Groofle is a digital marketing agency dedicated to helping brands find their creative voice and step into the spotlight. The agency works with businesses across sectors to build strategic roadmaps, craft compelling brand narratives, and deliver measurable growth through targeted digital initiatives. With a strong belief in the power of storytelling, Groofle champions brands by leveraging their unique strengths and translating them into high-impact digital experiences.

Groofle's portfolio spans clients in the hospitality, lifestyle, and luxury segments including brands such as The Countryside Resorts, Arison Luxury, and Molly's Café, reflecting the agency's versatility and creative range.

INTERNSHIP DETAILS

Name	Syed Faisal Andrabi
Role	Data Analyst
Organisation	Groofle
Department	Data & Analytics
Duration	3 Months
Nature of Work	Data Analysis, Performance Reporting, Campaign Analytics, Insights & Strategy

ABOUT THE ROLE

As a Data Analyst Intern at Groofle, the role centred on transforming raw marketing and campaign data into actionable insights that directly informed client strategy. Working within the Data & Analytics function, the internship involved close collaboration with the creative and account management teams to ensure that data-driven thinking was embedded across every stage of campaign planning, execution, and review. The role provided hands-on exposure to

the analytical tools and methodologies used by a modern digital marketing agency operating across multiple client verticals.

ROLES & RESPONSIBILITIES

- Collected, cleaned, and structured data from social media platforms, paid advertising dashboards, and web analytics tools including Google Analytics, Meta Ads Manager, and LinkedIn Campaign Manager
- Built and maintained performance dashboards and weekly reporting templates to track KPIs such as reach, engagement rate, CTR, ROAS, and conversion metrics across client accounts
- Conducted audience analysis and segmentation to identify high-performing demographics and content formats for targeted campaigns
- Performed competitive benchmarking and trend analysis to support strategic planning and content calendar decisions
- Analysed influencer marketing campaign data to evaluate ROI and inform future influencer selection and partnership strategy
- Assisted in A/B testing initiatives for paid social campaigns, interpreting results and presenting findings to account teams
- Prepared data visualisations and insight decks for internal reviews and client presentations
- Supported the team in identifying patterns and anomalies in campaign performance, recommending optimisations based on findings

KEY CONTRIBUTIONS & LEARNINGS

The internship at Groofle offered a rigorous, practical grounding in marketing analytics within a live agency environment. Working across multiple client accounts provided exposure to diverse data landscapes, each with its own performance benchmarks, audience behaviours, and strategic objectives which significantly sharpened the ability to contextualise data and draw meaningful conclusions rather than simply report numbers.

A key learning was the importance of translating complex data into clear, compelling narratives for non-technical stakeholders. Building dashboards and presenting insights to account managers and clients developed both technical proficiency and communication skills which is an increasingly critical combination in data-driven roles.

The experience also reinforced the value of rigour and accuracy in analytical work, particularly in a client-facing environment where decisions hinge on the reliability of reported data. Overall, the internship at Groofle was a highly formative experience that bridged academic foundations in data analysis with the practical realities of working in a fast-paced, creative industry which in turn helped me in building a strong base for a career in data analytics and business intelligence.

Appendix

Internship Certificate

GROOFLE

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Abrar Yaseen
CEO, Groofle Media Pvt. Ltd.

May 15, 2026

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Syed Faisal Andrabi** worked as an Intern at **Groofle**, a Digital Marketing Agency, within the **Data & Analytics** department, during the period from **February 15, 2026 to May 15, 2026**.

During this internship, Mr. Faisal demonstrated commendable dedication, analytical acumen, and a collaborative spirit. He contributed meaningfully to campaign performance analysis, dashboard development, audience segmentation, and data-driven strategic reporting across multiple client accounts within the Data & Analytics function. We wish him the very best in his future endeavors.

Yours sincerely,

Abrar Yaseen Mattoo

Abrar Yaseen Mattoo
CEO, Groofle Media Pvt. Ltd.

Plagiarism Report

Copyleaks

Analysis Report

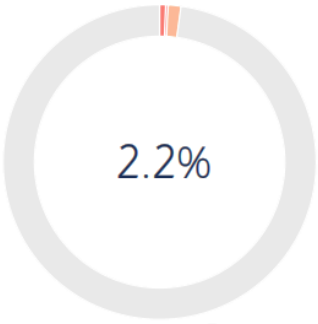
Plagiarism Detection and AI Detection Report

Syed Faisal Andrabi MSc BFA (H) (Sem 4) (1).pdf

Scan details

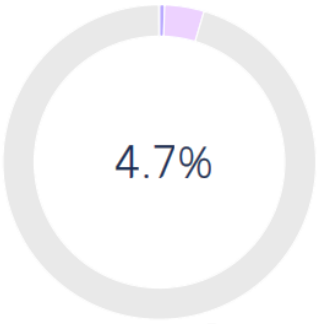
Scan Time	Total Pages	Total Words
May 16th, 2026 at 07:21 UTC	75	18566

Plagiarism Detection



Plagiarism types	Text coverage	Words
Identical	0.6%	103
Minor Changes	0.3%	43
Paraphrased	1.3%	218
Excluded		
Omitted Words		1

AI Detection



	Text coverage	Words
AI Text	4.7%	777
Low Frequency		89
Medium Frequency		6
High Frequency		6
Human Text	95.3%	15
Excluded		
Omitted Words		1